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BRAD DEVICES

Built in Africa, for the World

BRAD DEVICES — Complete Unified Specification

A full-stack silicon/hardware/software universe by The Architect Documents span July 2025 – July 2026. This spec book consolidates all 79 reference `.md` docs into one document. Last

updated: September 2026

How to Read This Spec

Generation Badges appear throughout to denote technology readiness:

Badge	Generation	Years	Meaning
[Gen1]	Kinetic	2026–2029	Shipping now. In production or ready for tape-out.
[Gen2]	Synaptic	2030–2033	Stretch goal. Plausible with process shrink and iteration.
[Gen3]	Quantum	2034–2037	Long-term R&D. Requires materials/physics breakthroughs.
[Gen4]	[Vision]	2038–2041	Deep materials research, no known production path.
[Gen5]	Singularity	2042+	Theoretical. Self-evolving gates, quantum fabric, bio-synaptic.
[Aesthetic]	—	—	Marketing language, not technical claims.

Release cadence: Brad Devices ships on a **4-year generation cycle** — each generation is a complete architectural leap, not an annual refresh. Within a generation, mid-cycle SKUs (process shrinks, memory/capacity upgrades, new form factors) are allowed, but the core architecture stays stable for the full 4-year window. Rationale and detail: `ROADMAP.md` §Release Cadence.

Cross-references in `code font` point to the source `.md` file for deeper detail.

Credibility tiers by document area: NPU, RAM, Drive, OS, Console, Datacenter (BradApex), BradXon platform = grounded engineering. Quantum Fabric = visionary. BradSense Render is the split case: the *architecture* (§2.5) is visionary, the *software reference upscaler* behind it now has a shipped M0 with receipts — see §2.5 and `Brad Silicon/GPU/BRADSENSE_RENDERER_PLAN.md`. (BradXon's ANF unit-morph is the one [Vision] shelf — see §1.5.)

Brand Pillar: Built in Africa, for the World

Brad Devices is positioned as **Africa's first vertically-integrated technology company** — custom silicon, operating system, and consumer devices designed, engineered, and integrated under one roof, on the continent. The claim is scoped precisely: no African company today designs its own CPU/GPU silicon, its own OS, and its own devices at integrated scale.

The Five Supporting Pillars

1. **Founder story** — rooted in Kisian; authenticity no competitor can copy.
2. **African engineering hubs** — Johannesburg data center (strategic, BradLabs, `Brad Software/OS/BMS_SPEC.md`), with Lagos/Nairobi/Cairo talent hubs planned; African engineers build the silicon, not just consume it.

3. **BradLabs foundry ambition** — a captive fab headquartered in Africa (2029+). Honest interim: **Samsung Foundry for all of Kinetic**, unified on a tiered node ladder (7LPP budget → SF4 → SF4P → SF4X flagship — cost + memory bundling), with a BradLabs custom PDK. Full tiering: `BradLabs/BRADLABS_FAB_STRATEGY.md`.
4. **Efficiency-first design** — low-power-first engineering (EROE, Falcon E-cores, DMIPS/W leadership) because power infrastructure is scarce across much of Africa — authentic to the continent *and* a product advantage everywhere.
5. **Developer ecosystem** — BradLabs Program (African game developers), 70/30 publishing split, BradStudios Originals (sibling company under the BradVerse Kenyan parent, not a Brad Devices division).

Honesty Guardrails

- The claim is **integrated scope only** — not "Africa's first tech company," which would be false.
- The fab is partner-run (**Samsung unified across all Kinetic tiers**, Intel optional US hedge — SMIC excluded) until BradLabs reaches 2029+; no premature "made in Africa" silicon claims.
- Every pillar has a concrete program behind it, never a slogan.

Existing Africa References (consolidated)

- `Brad Software/OS/BMS_SPEC.md:123` — Johannesburg data center (strategic, BradLabs)
- `BRAD_DEVICES_COMPLETE_SPEC.md` §8.1, §9.3 — BradLabs Program (African devs)
- `Brad Devices/Console/BRADAXIS_SPEC.md` — African-centric developer pillar

PART 0: ECOSYSTEM BLUEPRINT

Directory Structure

```
Brad Devices/
├── BRAD_DEVICES_COMPLETE_SPEC.md    ← You are here
├── MASTER_INDEX.md                  ← Ecosystem blueprint (375 lines)
├── VISION.md                        ← Sci-fi catalog with badges (172 lines)
├── ARCHITECTURE.md                  ← ASCII diagrams (196 lines)
├── ROADMAP.md                       ← Chronological evolution (240 lines)
├── NAMING_GUIDE.md                  ← Naming resolutions (327 lines)
├── BRAD_VS_APPLE.md                 ← Apple comparison (167 lines)
├── HISTORY.md                       ← Shipped-receipts ledger (235 lines)
├── TASKS.md                         ← The live furnace (90 lines)
├── AGENT.md                         ← Agent rulebook (447 lines)
├── README.md                        ← Repo guide (57 lines)
├── Brad Devices/
│   ├── BRADCUBE_SPEC.md             ← BradCube mini-PC line (105 lines)
│   ├── VR and BradSkynet/BRADSKYNET_ASTRA_LABS.md ← Horizon concepts (170 lines)
│   ├── Display/SENTINELVIEW.md      ← Display tech (157 lines)
│   ├── Console/BRADAXIS_SPEC.md     ← Console spec (305 lines)
│   ├── Storage/BRADDRIVE_COMPRESS_ARCH.md ← Storage architecture (186 lines)
│   └── Vision/BRADVISION_COMPLETE.md ← Camera & vision system (398 lines)
├── BradLabs/
│   └── BRADLABS_FAB_STRATEGY.md     ← Foundry & process ladder (87 lines)
├── Brad Silicon/
│   ├── BRADSILICON_STRATEGY.md      ← Silicon strategy (194 lines)
│   ├── PLATFORM_SPEC.md             ← BradBoard + SoC platforms (311 lines)
│   ├── PIPELINE_SPEC.md             ← Data pipeline (263 lines)
│   ├── Platform/BRADBOARD_DEEP_DIVE.md ← PCB stackups, Fabric, HSM (638 lines)
│   ├── BradISA/
│   │   ├── ARCHITECTURE_GUIDE.md    ← BradISA overview (105 lines)
```

```
│   │   ├── ARCHITECTURE_REFERENCE_MODEL.md ← Reference model (93 lines)
│   │   ├── ISA_QUICK_REFERENCE.md   ← Encode/decode tables (287 lines)
│   │   ├── BRADISA_VS_RISCV.md      ← ISA vs RISC-V + the six fences (156 lines)
│   │   ├── TOOLCHAIN_GUIDE.md       ← bradasm/asm workflow (180 lines)
│   │   └── TOOLCHAIN_STRATEGY.md    ← Toolchain roadmap (150 lines)
│   ├── CPU/
│   │   ├── HYPERCORE_ATLAS.md       ← HyperCore mobile SoC (243 lines)
│   │   ├── HYPERCORE_API_REF.md     ← SoC API (289 lines)
│   │   ├── HYPERCORE_REGMAP.md      ← Register map (244 lines)
│   │   ├── BRADCHIP.md              ← SoC package (124 lines)
│   │   └── EROE.md                  ← Power/thermal governor (51 lines)
│   ├── Compute/
│   │   ├── BRADAPEX_ACCELERATOR_SPEC.md ← Datacenter AI & HPC accelerator (237 lines)
│   │   └── BRAD_COMPUTE_ARCHITECTURE.md ← BradFx/BradGfx + BradApex (311 lines)
│   ├── GPU/
│   │   ├── GPU_ARCH.md              ← GPU arch + die names/registry (311 lines)
│   │   ├── GPU_ISA.md               ← BradVector ISA (293 lines)
│   │   ├── GPU_COMPUTE.md           ← Vector compute model (331 lines)
│   │   ├── GPU_DRIVER.md            ← GPU driver suite (282 lines)
│   │   ├── GPU_MEMORY.md            ← GPU memory model (281 lines)
│   │   ├── GPU_REGMAP.md            ← Register map (264 lines)
│   │   ├── BRADSENSE_RENDER_SPEC.md ← BradSense-Render deep-dive (279 lines)
│   │   └── BRADSENSE_RENDERER_PLAN.md ← Reference-upscaler rung plan (148 lines)
│   ├── NPU/
│   │   └── BRADNPU_ARCH.md          ← NPU architecture (181 lines)
│   └── Memory/
│       ├── BRADRAM_SPMP_ARCH.md     ← Memory architecture (174 lines)
│       ├── BRADRAM_DDS.md           ← Data sensing (234 lines)
│       ├── BRADRAM_POWER.md         ← Power model (191 lines)
│       ├── BRADRAM_RAS.md           ← Reliability (220 lines)
│       ├── BRADRAM_TIMING.md        ← Timing (187 lines)
│       ├── BRADRAM_VMEM.md          ← Virtual memory (220 lines)
│       ├── BRADRAM_REGMAP.md        ← Register map (242 lines)
│       └── MVRAM_RESEARCH.md         ← MVRAM memory research (LPGDDR rejected)
├── Brad Software/
│   └── OS/
│       ├── BRADOS_COMPLETE.md       ← OS + BMS (226 lines)
│       ├── BRADOS_KERNEL_STRATEGY.md ← Kernel roadmap (167 lines)
│       ├── BMS_SPEC.md              ← Brad Mobile Services (664 lines)
│       ├── BRADCVT_DRIVERS.md       ← Drivers suite (205 lines)
│       ├── BRADVECTOR_PLATFORM_MOAT.md ← BradVector platform & moat (295 lines)
│       └── CUSTOM_TECH.md           ← Custom-stack tech notes (89 lines)
├── COMPANY/
│   ├── COMPANY_STRUCTURE.md         ← BradVerse Group → Brad Devices (116 lines)
│   ├── DIVISIONS.md                 ← Division map (22 lines)
│   ├── IP_REGISTRY.md               ← Unified IP registry (208 lines)
│   ├── EQUITY_PLAN.md               ← Founder-first cap table (104 lines)
│   ├── NDA_TEMPLATE.md              ← Non-disclosure agreement (157 lines)
│   └── POLICIES.md                  ← Company policies (212 lines)
├── Peripherals/                     ← [ARCHIVE] — originals + raw transcript
│   ├── ARCHIVE.txt                  ←   archive manifest (read first)
│  └── [proprietary transcript]    ←  AI co-architect transcript (archived, not published)
│   └── Additional Details.docx/.odt, *.jpg  ← source docs + legacy assets
├── src/                             ← Reference toolchain: BVRT, braddev, bvlibs,
│                                      bms, tifa, rtl + 8 READMEs (736 lines)
└── Converted/                       ← [ARCHIVE] — pre-conversion source exports
    ├── ARCHIVE.txt                  ←   archive manifest (read first)
```

```
└── 17 more .txt/.md            ←   old folder layout, superseded by live docs
```

Total ecosystem: 84 `.md` text files = 84 docs, 17,635 lines (excluding Converted/ originals; 86 files, 18,527 lines with them). New since prior revision: BradISA/ suite (ISA_QUICK_REFERENCE, BRADISA_VS_RISCV, TOOLCHAIN_GUIDE, TOOLCHAIN_STRATEGY), GPU ISA suite (GPU_ISA/COMPUTE/DRIVER/MEMORY/REGMAP), BradSense-Render pair (BRADSENSE_RENDER_SPEC + BRADSENSE_RENDERER_PLAN), Memory suite (DDS/POWER/RAS/TIMING/VMEM/REGMAP/PARTNERSHIP), COMPANY/, BradLabs/, BRADSILICON_STRATEGY, BRAD_COMPUTE_ARCHITECTURE, BRADVECTOR_PLATFORM_MOAT, BRADOS_KERNEL_STRATEGY, BRADCUBE_SPEC, SENTINELVIEW, HYPERCORE_API_REF/REGMAP, BRADBOARD_DEEP_DIVE. This spec book merges all dedicated docs.

Architecture Stack

```
APPLICATION   BradPhones | BradSlates | BradMatrix | BradOS Compute | BradAxis | BradSkynet
                    ↑              ↑           ↑            ↑            ↑           ↑
MOTHERBOARD    BradBoard Mobile | BradBoard Matrix | BradBoard Nexus | BradBoard Rack
                    ↑              ↑           ↑            ↑            ↑           ↑
PLATFORM          BradMind (Efficiency) ──────────── BradNeuro (Performance) ──────────
BradXon (Flagship)
                    ↑              ↑           ↑            ↑            ↑           ↑
FABRIC              BradFusion Fabric (electrical [Gen1] + optical [Gen2] + quantum-
entangled [Vision])
                    ↑              ↑           ↑            ↑            ↑           ↑
COMPONENTS    BradCore CPUs | BradFx/BradGfx GPUs | BradNPU | BradApex (datacenter) | BradRAM/SPMP | Br
```

The Defining Architectural Insight: The Data Pipeline

```
BradDrive → BradCompress HDB → SPMP → BradFusion Fabric → Any Processor
```

Zero-copy, zero-CPU-overhead, direct-to-pool streaming. This is the ecosystem's core differentiator — Apple-style vertical integration where every component talks directly to every other via the Fabric, with no CPU involvement in data movement.

Generation Naming

Brad Devices ships on a **4-year generation cycle** — a complete architectural leap per generation, with mid-cycle SKU/process refinements in between. See `ROADMAP.md` §Release Cadence for full rationale.

```
Gen 1 (2026–2029):  KINETIC   — "Data in Motion. Intelligence in Real-Time."
Gen 2 (2030–2033):  SYNAPTIC  — "Learning at the Speed of Thought."
Gen 3 (2034–2037):  QUANTUM   — "Beyond Binary. Into Entanglement."
Gen 4 (2038–2041):  [VISION]  — Experimental node (BradLabs, sub-1nm).
Gen 5 (2042+):      SINGULARITY — Theoretical (self-evolving, quantum fabric).
```

> **Note on BradApex timing:** BradApex is a standalone datacenter accelerator, not a consumer generation. Its [Gen1] design → [Gen2] ramp 2027–28 occurs *within* the Gen1 Kinetic window (2026–2029) as a product ramp, complementing the consumer 4-year cadence.

PART 1: CPU ARCHITECTURE

1.1 CPU Philosophy: Two Families (+ Datacenter Engine)

	BradCore	BradXon	BradApex
Role	Ubiquitous workhorse — mass-market	Bleeding-edge flagship — apex predator	Datacenter engine — AI & HPC accelerator
Markets	Phones, tablets, laptops, compute, automotive, health, education, BradAxis console (standard SKU)	VR ecosystem servers (BradAstra), BradOS Compute Pro, BradSkynet controller	BradSkynet nodes (AI training/inference + scientific FP64)
Architecture	BradISA V1 custom P/E cores	BradISA P/E cores + BradGfx Max + Neuro-Stream; ANF = compute partitions + reconfigurable NPU + fabric governor	BA-ISA: tensor engine (FP8/FP16/BF16/INT8) + first-class IEEE FP64 vector engine
Process	Samsung 7LPP→SF4X tiered ladder [Gen1]	SF4X (4nm-class) chiplet [Gen1]; sub-1nm graphene/BN [Vision]	SF4X (4nm-class) [Gen2 ramp 2027–28]
GPU pairing	BradFx (integrated)	BradGfx Max (integrated via ANF)	None (not a GPU; memory-centric by design)
Node	`Brad Silicon/CPU/BRADCHIP.md`	`Brad Silicon/PLATFORM_SPEC.md`	`Brad Silicon/Compute/BRADAPEX_ACCELERAT`

1.2 BradCore — The Ubiquitous Workhorse [Gen1]

BradCore is the mass-market CPU core complex — asymmetric BradISA V1 with Phoenix (P) performance cores and Falcon (E) efficiency cores. Fabricated on the Samsung tiered ladder (7LPP→SF4X) at Samsung (unified fab; see `BradLabs/BRADLABS_FAB_STRATEGY.md`).

Core types: - **Phoenix (P):** High-performance out-of-order cores for compute-heavy workloads - **Falcon (E):** Power-efficient in-order cores for background tasks and system maintenance

Integration: BradCore ships as part of the BradChip SoC package (see 1.4), paired with BradFx GPU and BradNPU.

1.3 HyperCore & Atlas Architecture — Mobile SoC [Gen1]

HyperCore is the mobile silicon line — purpose-built for the pocket, not a cut-down PC. Targets: BradPhones and BradSlates.

Philosophy: BradCore is for sustained throughput. HyperCore is for **context persistence** — seamless app switching, instant resume, aggressive power gating.

	BradCore (Desktop)	HyperCore (Mobile)
Primary Goal	Throughput (speed)	Context (persistence)
UMA Style	SPMP — wide-bus, high-bandwidth	AUC — state persistence, instant intelligence
Compute Logic	Phoenix & Falcon	Phoenix & Sentinel
GPU Engine	BradFx / BradGfx	HyperCore GPU (Atlas-tuned)
Memory Buffer	2GB SRAM (Gfx Max)	950MB MVRAM (LPDDR5X)
Cooling	Active micro-fluidics	BradCool-Nano passive vapor
Node	4nm→2nm→sub-1nm	7LPP/SF4 (mature, stable)

HyperCore Product Lineup

Tier	Model	NPU	MVRAM	Target	Key Advantage
Lite	HC-100	16-core	128MB	BradPhone Essential	3-day battery, GTX 1050-class GPU
Standard	HC-300	32-core	256MB	BradPhone Base	Balanced AI-agent performance
Ultra	HC-500	48-core	450MB	BradPhone Ultra	Atlas Pro engine
Elite	HC-Elite	64-core	950MB	BradSlate Prime	Matches MacBook M2-class performance

Atlas Architecture Components

- **Atlas Unified Context (AUC):** Shared context, not just shared memory. CPU and GPU share context with zero-latency context switch (<100µs). GPU power-gates completely during text reading.
- **Sentinel Cores:** Ultra-low-leakage always-on cores (microwatts, not milliwatts) — distinct from Falcon cores. Handle clock, notifications, TIFA AI listening, FaceLock. Enable 48h+ standby.
- **MVRAM (Management Video RAM):** Dedicated on-die buffer using LPDDR5X, 3D-stacked tile on logic. Max 950MB on HC-Elite, 256 GB/s bandwidth, ~500mW active / ~50µW standby. Permanently caches: BradOS kernel, TIFA AI model weights, UI asset library, frozen app contexts (10+ apps).
- **HyperCore GPU:** Same BradVector Engines and Neuro-Stream Units as desktop BradFx/Gfx. Same shader math — desktop code runs on mobile without port. Scaled core count, same architecture.
- **BradCool-Nano:** 0.2mm vapor-chamber/graphene sandwich. Peak <45°C surface sustained gaming.

Competitive Positioning (2026)

Metric	Apple A19 Pro	HyperCore HC-Elite
Node	3nm (N3P)	SF4X (4nm-class)
On-Die Buffer	~32MB System Cache	950MB MVRAM (LPDDR5X)
Architecture	Legacy hybrid	Atlas Unified Context
AI Focus	Photography / Siri	Agentic reasoning (TIFA)
Battery	18-24 hours	36-48 hours (Sentinel)

1.4 BradChip — Unified SoC Package [Gen1]

BradChip combines BradCore CPU + BradFx GPU + BradNPU + memory controller + BradFusion Fabric into one monolithic or chiplet-based die. It is the compute module that powers BradMatrix, BradSlates, and BradOS Compute Base systems. Slots into BradBoard Matrix socket.

Tier	CPU	GPU	NPU	Fabric	Process	TDP	Target
Lite	P2+E4	BradFx 10	256 NCC	Fabric	7LPP (7nm)	10-15W	BradMatrix Essential
Standard	P4+E4	BradFx 30	384 NCC	Fabric	SF4 (4nm)	15-25W	BradMatrix Base, BradSlate Prime
Ultra	P6+E4	BradFx 50	512 NCC	Fabric + optical	SF4P (4nm)	25-40W	BradMatrix Prime 14”

Tier	CPU	GPU	NPU	Fabric	Process	TDP	Target
Elite	P8+E4	BradFx 70	768 NCC	Fabric + optical	SF4X (4nm)	40-65W	BradMatrix Ultra 16”
Max	P8+E8	BradFx 90 (Max-i)	1024 NCC	Fabric + optical	SF4X (4nm)	65-95W	BradOS Compute Base

All tiers include: BradFusion Fabric, dual-channel memory controller (Brad-CAMM LPDDR/DDR), BradSec root-of-trust, BradCompress HDB interface, BradLink tethering.

BGA/LGA packages ranging from 25×30mm (Lite, soldered) to 45×50mm (Max, LGA socket + backplate).

Competitive comparison with Apple M4 Max — BradChip Max offers similar CPU config (16-core vs 16-core), competitive GPU (48 VE vs 40-core), stronger NPU (1024 NCC + HB-AIM vs 38 TOPS), upgradable Brad-CAMM LPDDR/DDR memory, and BradFusion Fabric + optical interconnect.

1.5 BradXon — The Flagship Adaptive Processor

BradXon is the flagship processor — a standalone chiplet CPU (not a mobile SoC) that integrates directly into BradBoard Nexus and Rack over the BradFusion fabric. Its defining feature is the **Adaptive Neural Fabric (ANF)**, Gen 1 [DESIGNED].

ANF — what “adaptive” means [Gen1, DESIGNED]: - **Compute Partitions (CPs):** the die’s execution arrays are grouped into secure hardware partitions (CPU / GPU / NPU sides), composed at boot under zone isolation (MIG-class). Partitions are re-allocated at domain boundaries, never per instruction. - **Reconfigurable Dataflow Array (RDA):** the NPU slice is a CGRA-class dataflow fabric, rebuilt at kernel launch — the AI math adapts to the model rather than to a fixed systolic pipe. - **Fabric Governor (FG):** the PMU + OS scheduler continually re-allocate clock, power, and fabric bandwidth between the CPU/GPU/NPU partitions (DVFS + heterogeneous scheduling). Budget reallocation is real; silicon reallocation is not.

ANF — what Gen 2 keeps as [Vision]: - **Unit morphing:** re-forming execution-unit tiles into a different domain category at runtime. The honest path is tile composition at integration: fabric tiles floorplanned CPU/GPU/NPU, mated into a package at build time — not per-instruction. - **Self-evolving logic gates:** [Vision] transistors that physically re-wire — requires fundamentally new device physics. - **Photonic / quantum fabric:** hybrid optical chiplet links [Gen2]; on-die photonic logic [Vision]; quantum-entangled pathways [Vision] — pure fiction by 2026 standards.

Integrated & companion silicon: - **Integrated BradGfx Max:** [Gen1] same silicon as the discrete card, attached via the ANF compute-partition layer - **BradGfx Max Xntensive:** [Gen1] discrete edition — 256 BradVector / 128 RT / 512 Neuro-Stream, 32 GB GDDR7 on a 192–512-bit bus, hybrid-UMA (dedicated VRAM + SPMP overflow) - **NeuroSense X-Units:** [Gen1] dedicated AIMP (AI Model Processors) - **BradSec-X Quantum Fortification:** [Vision] post-quantum crypto with self-healing integrity

Targets: BradSkynet Core Nodes, BradAstra Servers, BradOS Compute Pro Workstations. *(BradAxis uses standard BradCore — see §8.1.)*

PART 2: GPU ARCHITECTURE

2.1 Product Lineup

BradGfx — Discrete-Package, Unified-Memory GPUs

> *BradGfx is a **UMA GPU** for Lite→Max (discrete package, unified SPMP memory, Grace-Hopper class — no private VRAM). Only **Max Xntensive** carries a GDDR7 tank and is the line’s one true **hybrid dGPU**. See `Brad Silicon/GPU/GPU_ARCH.md` §1.*

SKU	Shader Cores	RT Cores	AI Cores	FP32 TFLOPS	TDP	Target
Lite	512	16	—	4-6	75W	Budget gaming
Standard	1024	32	8	10-14	130W	Mainstream (~RTX 3050/3060)
Ultra	2048	64	16	20-28	220W	Prosumer
Max	4096	128	32	40-55	350W	Flagship (discrete-package UMA or ANF-integrated)
Max Xn-ten-sive	4096	128	32+	40-55+	225W	Flagship hybrid dGPU — GDDR7 tank + SPMP (see GPU_ARCH §1)

BradFx — Integrated GPUs

SKU	Shader Cores	RT Cores	AI Cores	FP32 TFLOPS	TDP	Target
Lite	128	—	—	1-2	5-10W	Budget devices
Standard	256	—	4	2-4	10-20W	Laptops/tablets (~Apple M2 GPU)
Ultra	512	16	8	5-8	20-35W	Premium ultrabooks
Max (Max-i)	1024	32	16	10-15	35-65W	BradChip, powerful slates

2.1a BradVector Engines — ISA & Compute Reference Docs

The full instruction set, compute model, and verification story for the BradVector engines (all shader cores above) live in the reference documentation in `Brad Silicon/GPU/` and `Brad Silicon/Compute/`:

Reference Doc	Covers
`Brad Silicon/GPU/GPU_ARCH.md`	Torox G1 microarchitecture — BradVector / BradRT / Neuro-Stream units, full SKU lineup, ray-tracing cores, **die naming (BGT/BFT/BAT) + Torox G1 die registry**
`Brad Silicon/GPU/GPU_ISA.md`	BradVector ISA — 64-bit dual-issue SIMT encoding, `.bvbc` portable bytecode vs `.brsh` native AOT, register file, shared memory

Reference Doc	Covers
`Brad Silicon/GPU/GPU_COMPUTE.md`	Compute model, kernel launch, warp scheduling, memory model (SPMP / Infinity Buffer overlap)
`Brad Silicon/GPU/GPU_REGMAP.md`	Register map & programming model — register-level control of all Bradley tiers
`Brad Silicon/GPU/GPU_MEMORY.md`	Memory model & cache hierarchy — SPMP access, coherency, bandwidth allocation
`Brad Silicon/GPU/GPU_DRIVER.md`	The GPU half of the driver layer — Vulkan 1.4, DirectX 12 Ultimate, Metal 3, Brad-Link API
`Brad Silicon/GPU/BRADSENSE_RENDER_SPEC.md`	BradSense-Render neural reconstruction deep-dive (see §2.5)
`Brad Silicon/GPU/BRADSENSE_RENDERER_PLAN.md`	The software reference upscaler — M0–M4 rungs, acceptance gates, receipt chain
`Brad Silicon/Compute/BRAD_COMPUTE_ARCHITECTURE.md`	BradVector as the CUDA-analog software platform; SPMP → CUDA-shared-equivalent, Infinity Buffer → CUDA-shared

Reference toolchain (released): `bradc` (assembler → `.bvbc`), BVRT (interpreter), `bradgdb` (debugger), BradTimeline (profiler/trace). Full details in `Brad Software/OS/BRADVECTOR_PLATFORM_MOAT.md`.

2.2 GPU Architecture Diagram [Gen1]

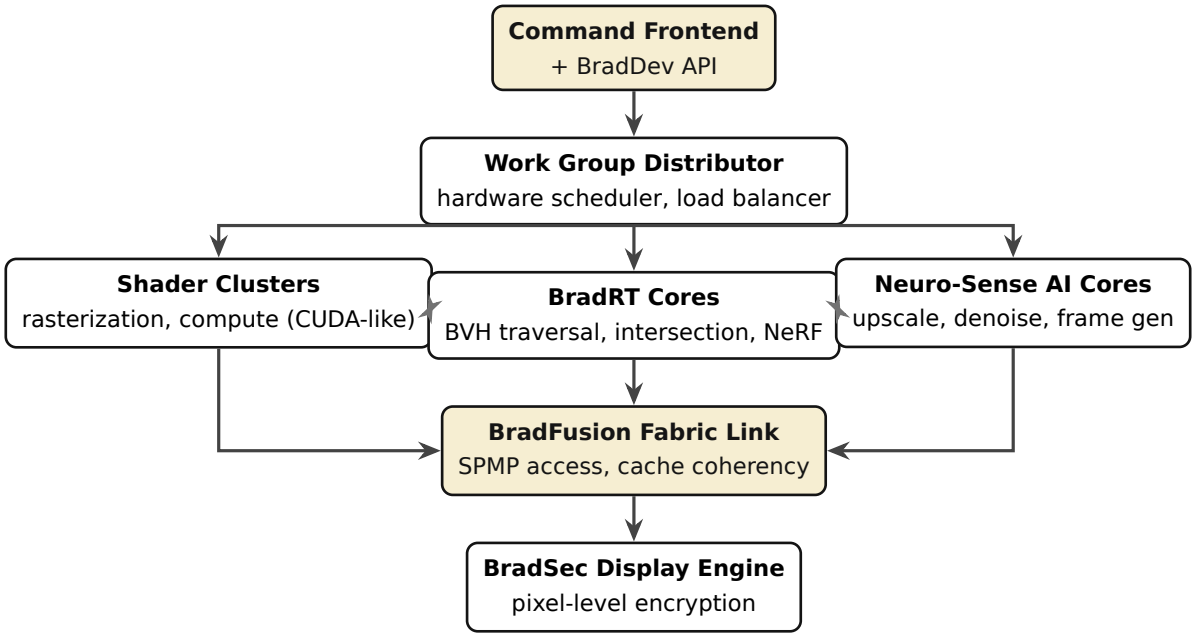


Figure 1: Gen1 unified GPU architecture — BRADGFX / BRADFX

2.3 BradRT Cores — Ray Tracing [Gen1]

Each BradRT Core contains: - **Ray Traversal Unit (RTU):** Navigates BVH acceleration structures with proprietary cache-optimized tree traversal - **Intersection Processing Unit (IPU):** Massively parallel ray-primitive intersection (triangles, spheres, line segments) - **Programmable Ray Pipeline (PRP):** Custom ray behaviors, material interactions, secondary ray gen — supports Opacity Micromaps - **Triangle Cluster Accelerator:** Optimized ray tracing of complex clustered geometry

RT scaling: 16 cores / 4 GigaRays/s (Lite) → 128 cores / 40+ GigaRays/s (Max).

Neural Rendering Acceleration [Gen1]: Hardware optimized for NeRF (Neural Radiance Fields) evaluation — real-time digital doubles and volumetric effects (fog, smoke).

2.4 NeuroSense AI Engine (On-GPU) [Gen1]

Feature	Description
BradSense-Render	AI upscaling + frame generation. Context-aware predictive rendering, not just DLSS/FSR.
NeuroSense-Denoise	AI denoising for ray-traced output — fewer rays per pixel while maintaining quality.
TIFA AI	Real-time workload monitoring — dynamically adjusts clock, voltage, rendering technique.
AI Physics	Dedicated AI cores accelerate physics simulation, character behavior, environmental dynamics.

Division of labor: On-GPU AI cores handle real-time rendering inference (denoising, upscaling, frame gen). BradNPU handles heavy AI (model training, complex inference, NeuroSense assistant).

2.5 BradSense-Render — Neural Rendering Deep-Dive [Gen1]

BradSense-Render is not an upscaler — it is a **reconstruction engine** that uses the BradFusion Fabric to pull semantic object data from the game engine, then applies per-object neural weights to reconstruct each frame at higher-than-native fidelity.

The Intelligence Pipeline:

1. **Semantic Feed (via Fabric):** Game engine sends compact semantic packet — object ID map, material type (8 classes), depth buffer, motion vectors, albedo/normal — ~50MB/frame uncompressed, <5MB via BradCompress.
2. **Per-Object Neural Reconstruction:** Each object class has a dedicated lightweight neural network (expert). Loading expert weights from on-die SRAM in <1µs per type switch.
3. **Temporal Stabilization:** Per-object accumulation rates — static objects accumulate 16+ frames, fast-moving objects 1-2 frames. Zero ghosting (system knows *which* pixels belong to *which* object).
4. **Frame Generation:** Interpolated frames between real renders. 1 real → 1 AI (2×) or 1:2 (3×). No input lag increase (generated retroactively between real frames).

Neuro-Stream Units (hardware requirements):

Tier	Neuro-Stream Units	On-die SRAM	Object Classes	Features
BradFx 10 (Lite)	32	8MB	4	Basic reconstruction

Tier	Neuro-Stream Units	On-die SRAM	Object Classes	Features
BradFx 30 (Standard)	64	16MB	6	—
BradFx 50 (Ultra)	96	24MB	8 (all)	—
BradFx 70 (Elite)	128	32MB	8 + temporal	Full
BradFx 90 (Max-i)	192	48MB	8 + frame gen	Full + FG
BradGfx 100 (Lite)	128	256MB	8 + FG + temporal	Full
BradGfx 300 (Standard)	256	512MB	8 + variable precision	Full
BradGfx 500 (Ultra)	384	1GB	8 + NeRF	Full
BradGfx 700 (Elite)	448	1.5GB	Full stack	Full
BradGfx Max	**192**	**2GB**	**Full + MoE**	**Full**
BradGfx Max Xntensive	**512**	**2GB**	**Full + MoE**	**Full**

Quality Modes: - **Quality:** 1440p→4K, 60% Neuro-Stream load - **Balanced:** 1080p→4K, 40% load (default) - **Performance:** 720p→4K, 25% load (competitive MP) - **Ultra Performance:** 540p→4K, 15% load (path tracing) - **Native AA:** 4K→4K, AI anti-alias only, 10% load

Frame Generation: 1× (double) or 2× (triple). No input lag penalty.

Software reference status (receipts, not projections): the pipeline above is the design intent. What runs today is the **M0 reference upscaler** in the `bradsense` driver — `bradsense_upscale_rgba()` (bilinear + Catmull-Rom bicubic, arbitrary ratio, luma-adaptive sharpen with an anti-ringing clamp), fed by `bradsense_supply_frame()` into `bradsense_reconstruct()`, which scales to the quality mode's target and returns `-ENODATA` rather than inventing pixels. Receipt: `src/tests/test_bradsense_pixels.c`, 17/17 `ctest` green under the CI configuration (`-DBRAD_BUILD_BVLIB=ON`; 12 in the default local build), 2× super-resolution PSNR 39.43 dB (bilinear) / 39.44 (bicubic) / 39.57 (bicubic + sharpen). Temporal accumulation (M1) is queued, the metrics corpus and golden CI gate are M2, BradVector kernel mapping is M3, and the trained per-object experts are a separate M4 track. No frame generation, expert inference, or throughput number is claimed as shipped. Rungs and gates: `Brad Silicon/GPU/BRADSENSE_RENDERER_PLAN.md`.

2.6 BradFusion Fabric Integration

All BradGfx and BradFx GPUs are direct native peers on the Fabric: - **Zero-copy access to SPMP:** No separate VRAM. GPU reads textures/meshes/shaders directly from the unified LPDDR pool. -

CPU-GPU-NPU coherency: Ultra-low-latency heterogeneous compute communication. - **Fabric bandwidth reservation:** 64 GB/s reserved for semantic data path (non-blocking, separate from pixel data).

2.7 BradGfx Max Dual Role

Both: discrete card (for BradOS Compute workstations) AND integrated into BradXon via ANF (for bleeding-edge flagship systems). Same silicon, different packaging.

2.7a BradGfx Max Xntensive (Hybrid-UMA Edition)

Discrete-only edition aimed at Windows Pros, workstations, and AI/MoE training. Dedicated **32 GB GDDR7** VRAM on a **192–512-bit bus** (0.75–2.0 TB/s) — acts as its own VRAM tank on Windows, and as a high-speed cache with SPMP overflow on BradOS (~100 GB+ effective). SF4X 4nm-class (Brad-Gfx Max tier), 225W nominal, ~98 TFLOPS FP32 / ~786 TFLOPS FP16 AI.

2.8 BradSec Display Pipeline [Gen1]

Feature	Description
Secure framebuffer	Encrypted render target, decrypted only at display output
Pixel-level encryption	HDCP 2.3+ with hardware root of trust
Asset protection	Textures/geometry stream encrypted from SPMP, decrypted on-GPU
Content tamper prevention	Integrity checks on display pipeline

---

PART 3: NPU ARCHITECTURE

3.1 BradNPU — Neural Processing Unit [Gen1]

BradNPU is the dedicated AI accelerator powering NeuroSense AI across the entire ecosystem. Purpose-built from the bottom up for neural workloads — matrix multiply, convolution, transformer attention, and mixture-of-experts routing.

Tier	Purpose	Deployment
BradCore + BradNPU	Pervasive AI everywhere	BradChip (Lite→Max), HyperCore (HC-100→Elite)
BradXon + X-Units	Bleeding-edge AI training	BradSkynet, BradAstra, BradOS Compute Pro
Standalone BradNPU	Edge AI acceleration	BradEco sensors, BradVision ISP co-processor

3.2 Neural Compute Cluster (NCC) Architecture

Each NCC is a **512-wide SIMD processor**:

Sub-Unit	Width	Function
Systolic array (INT4/INT8)	512×512	Matrix multiply, convolution (inference)
SIMD vector (FP16/BF16)	32-wide	Activations, layer norm, softmax
SIMD vector (FP32)	16-wide	Training, high-precision fallback

Sub-Unit	Width	Function
Sparse engine	—	2:4 structured pruning — skip zero weights at wire speed
Aggregation buffer	256 KB	Local reduce/scatter for attention, MoE routing

NCC Cluster Topology:

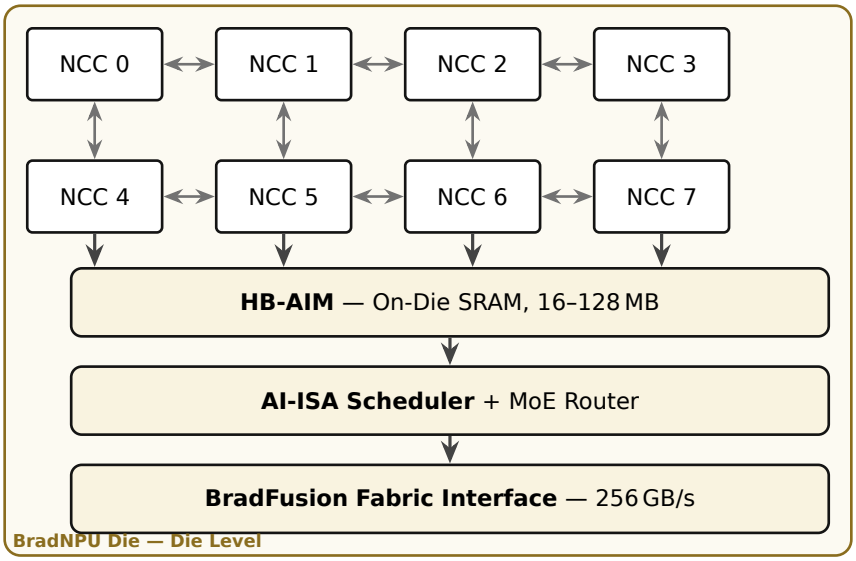


Figure 2: NCC cluster topology — 2×4 NCC mesh feeding HB-AIM, the AI-ISA scheduler, and the fabric

NCC counts by tier:

Tier	NCCs	INT8 TOPS	FP16 TFLOPS	HB-AIM	TDP (NPU only)
BradChip Lite	16	32	8	16 MB	1.5 W
BradChip Standard	24	48	12	32 MB	2.5 W
BradChip Ultra	32	64	16	64 MB	4 W
BradChip Elite	48	96	24	128 MB	6 W
BradChip Max	64	128	32	256 MB	9 W
HyperCore HC-300	8	16	4	8 MB	0.8 W
HyperCore Elite	24	48	12	32 MB	2.5 W

3.3 HB-AIM (High-Bandwidth AI Memory)

On-die SRAM dedicated to model weights and activations: - **Type:** Custom 6T SRAM (high-density, low-voltage) - **Bandwidth:** 2 TB/s per 16 MB tile (512-bit @ 2 GHz) - **Tiling:** 8×8 MB tiles per 64 MB region, independently powered - **Retention:** Near-threshold mode (0.4 V) — 5 µW/MB standby - **MoE spooler:** When experts exceed HB-AIM, cold experts spilled to SPMP via Fabric. MoE Router prefetches next N experts based on attention pattern prediction.

3.4 AI-ISA (Dedicated Instruction Set)

Custom single-cycle instructions on NCC hardware:

Category	Instructions
Matrix	`MATMUL`, `CONV`, `BATCHED_GEMM`
Activation	`RELU`, `GELU`, `SILU`, `TANH`, `SIGMOID`
Normalization	`LAYER_NORM`, `BATCH_NORM`, `RMS_NORM` (4 cycles fused)
Attention	`QKV_PROJ`, `ATTN_MASK`, `SOFTMAX`, `ATTN_OUT` (6 cycles fused)
MoE	`TOP_K_ROUTER`, `EXPERT_COMBINE`, `SPARSE_GATHER`
Memory	`HB_LOAD`, `HB_STORE`, `HB_SCATTER`, `HB_GATHER`
Sparse	`PRUNE_MASK`, `SPARSE_MATMUL`, `DENSE_RECOMPOSE`
Quant	`FP16_TO_INT8`, `INT8_TO_FP16`, `FP16_TO_FP4`

Compiler target: PyTorch/TensorFlow models graph-rewritten to AI-ISA.

3.5 MoE (Mixture of Experts) Engine

- **Top-2 routing:** Each token selects 2 of N experts
- **Router capacity:** 1024 experts per NCC cluster, 8192 per die
- **Expert size:** 1M–500M parameters (configurable)
- **Prefetch pipeline:** Router runs N tokens ahead of compute — expert prefetch overlaps with current inference, hiding latency

3.6 Workload Scheduler

Priority	Workload	Latency Budget	NCC Allocation
Critical	FaceLock, voice auth	<10ms	4 NCCs dedicated
High	BradVision Fusion, real-time AI	<50ms	8-16 NCCs
Medium	TIFA inference, BradCoach	<200ms	Remaining NCCs
Low	Background indexing, predictive text	>500ms	1-4 NCCs (low power)
Bulk	Offline training	Best effort	All idle NCCs

3.7 Use Case Performance

Use Case	Model	NCCs	Latency
Face unlock	MobileFaceNet	2	3ms
Real-time depth (BradVision)	MiDaS v3 small	4	8ms @ 60fps
Speech-to-text	Whisper tiny	4	12ms per 5s audio
TIFA chat (7B model)	LLaMA-7B 4-bit	48	25 tokens/ms
SDXL image gen	Stable Diffusion XL	64	2s per 1024×1024

3.8 Fabrication & Power

Parameter	Specification
Process	7LPP (Gen1 budget), SF4/SF4P (Gen1)
Frequency	1.2 GHz (turbo), 400 MHz (efficient), 50 MHz (always-on)

Parameter	Specification
Power states	P0-P5 + OFF with 5µs wake latency
Voltage	0.4 V (retention) – 0.9 V (turbo)
Always-on domain	2 MB SRAM + lightweight scheduler — 2 mW

PART 4: MEMORY & STORAGE

4.1 Memory Architecture: BradRAM + SPMP [Gen1]

Two-tier unified architecture: - **BradRAM** = the brand for our memory subsystem: standard JEDEC LPDDR5X/DDR5 dies + a Brad-designed controller (DDS predictive prefetch, L4 SRAM system-level cache, inline encryption, EROE telemetry) - **SPMP** = logical unified pool formed by the BradRAM subsystem + NeuroSense AI controller + Fabric integration

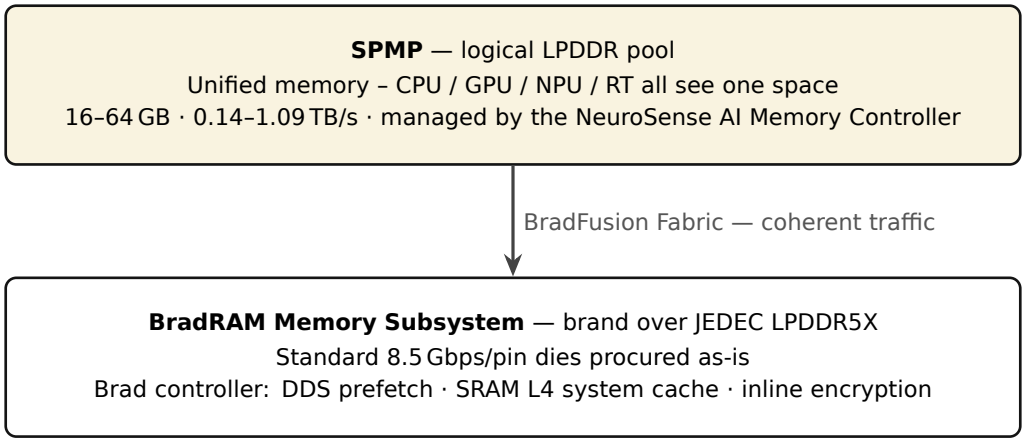


Figure 3: BradRAM + SPMP — two-tier unified memory: logical pool above the subsystem, joined by the fabric

BradRAM Subsystem Specs

Parameter	Value
Base standard	JEDEC LPDDR5X-8533 (JESD209-5B), off-the-shelf dies
Pin speed	8.5 Gbps per pin (Gen1) / 14.4 Gbps (Gen2 LPDDR6)
Bus width	16-bit per die (128–1024-bit system bus)
Bandwidth per die	17 GB/s (LPDDR5X) / 38.4 GB/s (LPDDR6)
Capacity per die	8 Gb (1 GB) – 32 Gb (4 GB)
Stack height	4–8 dies per stack (wide-bus UMA)
Stack capacity	4 GB – 32 GB per stack
Voltage	VDD 1.05 V (active), VDDQ 0.5 V, 0.6 V (retention)
Process	Industry DRAM process (dies) + Brad memory-controller die
Brad content	Controller + L4 SRAM + encryption — **no custom DRAM die**

Dynamic Data-Sensing (DDS) DDS is implemented on the **Brad memory-controller die** (0.5 mm², ~50 µW), not inside the DRAM (standard JEDEC parts cannot host custom logic): - Monitors access patterns from CPU, GPU, NPU - Learns per-application access signatures - Proactively prefetches from BradDrive into L4 SRAM cache - Communicates prefetch hints to Drive controller via Fabric

Effectiveness: 85% hit rate for predicted workloads. Reduces SPMP demand by 30%.

Memory Hierarchy

Level	Type	Size per Stack	Access	Location
L4	SRAM (system-level cache)	64 MB	2 ns	On Brad memory-controller die
L3	LPDDR (fast)	25% of stack	10 ns	TSV-connected chiplets
L2	LPDDR (main)	75% of stack	20 ns	Stacked dies
SPMP	Logical pool	16–64 GB	20–40 ns (via Fabric)	Aggregated

The L4 SRAM is an Apple-SLC-style system-level cache on the controller die; the LPDDR tiers below it are standard JEDEC dies.

SPMP Configurations

Tier	Bus Width	Capacity	Bandwidth (LPDDR5X)	Bandwidth (LPDDR6)	Fabric Paths	Used In
Lite	128-bit	8–16 GB	136 GB/s	230 GB/s	2× 64-bit	BradMatrix Essential
Standard	256-bit	16–32 GB	272 GB/s	461 GB/s	4× 64-bit	BradMatrix Base, BradSlate Base
Pro	384-bit	32–48 GB	409 GB/s	691 GB/s	6× 64-bit	BradMatrix Prime
Max	512-bit	48–64 GB	545 GB/s	922 GB/s	8× 64-bit	BradOS Compute Pro
Enterprise	1024-bit	64–128 GB	1.09 TB/s	1.84 TB/s	16× 64-bit + optical	BradSkynet server

NeuroSense AI Memory Controller Dedicated RISC-V core cluster + neural accelerator between BradRAM and Fabric: - **Access prediction:** LSTM-based sequence model per active process (+5ns pipelined) - **Data placement:** Hot → L4 SRAM, warm → L3 fast LPDDR, cold → L2 main - **QoS arbitration:** Weighted fair queuing with urgency scoring - **Encryption:** AES-256-GCM inline (+3ns per 64B line) - **Integrity:** Merkle tree (SHA-3-256) verification concurrent with read

Unified access model:

```
Processor Read Request → BradFusion Fabric (1–4 cycles)
    → SPMP Controller
    ├── L4 SRAM hit → return (2 ns)
    ├── L3 fast LPDDR hit → return (10 ns)
    ├── L2 main LPDDR hit → return (20 ns)
    └── Miss → Fabric → BradDrive HDB → decompress → SPMP write → return (<1ms)
```

Zero-copy principle: Data written once. CPU reads pixel buffer, GPU renders from same address, NPU infers from same. No copies.

BradSec-RAM [Gen1] Hardware-level memory encryption in the **Brad memory controller data path** (between CPU/GPU/NPU and the standard JEDEC dies): - AES-256-GCM per memory channel, per-session keys from PUF root-of-trust - Per 64-byte cache line with unique IV - Merkle tree integrity verification - +3ns per access, +5% power (always-on, no bypass) - Key scrub on chassis-open detect

EROE Thermal Integration

- Per-die thermal sensors (0.1°C resolution) → BradOS EROE reads via Fabric
- Adaptive DRAM refresh rate (2× at 85°C+, 0.5× at <40°C)
- Hot page migration to cooler dies
- Bandwidth capping per application
- Sustain: 95% peak bandwidth indefinitely under 85°C

4.2 Storage Architecture: BradDrive + BradCompress [Gen1]

Data Pipeline:

```
Application → File Request
    ↓
BradDrive Controller (BDC)
    ├── NeuroSense-Storage AI predicts asset need
    ├── Read from BradNAND (compressed)
    └── Hardware Decompression Block (HDB)
        ↓ Decompressed (zero-CPU)
SPMP (unified memory pool) ← direct DMA, no RAM staging
    ↓
Any processor: CPU / GPU / NPU / RT
```

Key differentiator: CPU never touches compressed or decompressed data. Entire path from NAND to SPMP is hardware-accelerated.

BradDrive Physical Drive

Model	Form Factor	Capacity	NAND Type	Seq. Read	Seq. Read (compressed)	Target
BradDrive	Brad-M.3 (2230)	512 GB–2 TB	TLC	14 GB/s	28 GB/s (2:1)	BradMatrix Base, BradSlate Base
BradDrive Pro	Brad-M.3 (2280)	1 TB–4 TB	MLC	16 GB/s	40 GB/s (2.5:1)	BradMatrix Prime

Model	Form Factor	Capacity	NAND Type	Seq. Read	Seq. Read (compressed)	Target
BradDrive Elite	Brad-M.3 (2580)	2 TB–8 TB	SLC	18 GB/s	54 GB/s (3:1)	Workstation (console uses 1/2 TB TLC, see §8.1)
BradDrive Enter-prise	EDSFF E3	8 TB–32 TB	SLC/MLC hybrid	28 GB/s	84 GB/s (3:1)	BradSkynet server

Interface: PCIe 6.0 x4 (64 GB/s uni, 128 GB/s bi). Brad-M.3 form factor (12mm wide, keyed for BradDrive only).

BradDrive Controller (BDC) Multi-core RISC-V processor + dedicated accelerators: - **RISC-V core cluster:** 4× in-order @ 1 GHz — command processing, FTL, scheduling - **BradCompress HDB:** Fixed-function decompression — 64 GB/s throughput - **NAND controller:** 16 channels @ 2400 MT/s per channel — ONFI 5.1 - **NeuroSense AI:** Tiny neural net (0.5 TOPS) — learns access patterns, predicts pre-fetch - **DRAM cache:** 2 GB LPDDR5X for hot data + FTL mapping table - **PCIe 6.0 PHY:** x4 lanes @ 64 GT/s PAM4 - **BradSec AES-256:** Inline encrypt-on-write, decrypt-on-read - **ECC:** LDPC + RAID-like cross-die parity — corrects up to 2Kbit errors/sector

BradNAND [Gen1]

Type	Bits/Cell	Program	Read	Endurance	Use
SLC	1	20 µs	15 µs	100K P/E	Elite, Pro (Brad Soft-ware/OS/data)
MLC	2	40 µs	20 µs	30K P/E	Pro (bulk)
TLC	3	80 µs	25 µs	5K P/E	Mass-market
QLC	4	150 µs	35 µs	1K P/E	Cold storage

3D layer count: 238 layers (Gen1), 350 layers (Gen2 — Vision).

NeuroSense-Storage AI On-controller neural network predicting data access: - Input: File system metadata, application context, user behavior, time-of-day - Output: Probability scores for next-N-second block access - Pre-fetch: Predicted blocks loaded from NAND → HDB decompression → SPMP before request - Rise of the Streets example: Learns map coordinate (x,y) → triggers zone texture load → preloads 500 MB before player arrives

BradCompress Compression Codec Three-stage lossless compression:

```
Input Data Block
    ↓ Stage 1: Dictionary Coding (64 KB sliding window, domain-tuned)
    ↓ Stage 2: Run-Length Encoding (collapses repeated tokens, adaptive threshold 3+)
    ↓ Stage 3: Entropy Coding (range encoding, 8× 2D context models)
    Output: bitstream
```

Domain Profiles:

Profile	Avg Ratio	Decomp Speed	Use
Mesh	3.5:1	56 GB/s	3D vertex data, index buffers
Texture	2.8:1	62 GB/s	BCn textures, raw RGBA
Audio	2.2:1	48 GB/s	WAV, raw PCM, synths
Video	2.5:1	45 GB/s	Intra-frames, motion vectors
AI Model	2.0:1	64 GB/s	FP16/INT8 weights
General	1.8:1	35 GB/s	Binary blobs, archives
Mixed	2.5:1 avg	52 GB/s	Asset bundles (game levels)

Hardware Decompression Block (HDB)

Parameter	Spec
Architecture	256-wide SIMD pipeline (custom, not GPU)
Clock	1.2 GHz
Throughput	64 GB/s peak, 40 GB/s sustained mixed
Latency	128 cycles (first byte), pipelined
L0 cache	256 KB — dictionary tables + recently decoded data
Fabric interface	Dedicated SPMP DMA engine — 64 GB/s direct write

Direct-to-SPMP zero-copy API:

```
void* addr = bradDirectLoad("/game/city_block_05.bcm", SPMP_TIER_FAST);
// No malloc, no memcpy, no CPU-side decompression
```

Security

- **Data-at-rest:** AES-256-XTS — encrypts on NAND write, decrypts on read
- **Data-in-transit:** AES-256-GCM — Fabric packets encrypted between HDB and SPMP
- **Anti-rollback:** Monotonic counter in BradSec prevents firmware downgrade
- **Self-encrypting drive:** TCG Opal 2.0 compliant

PART 5: VISION & CAMERA

5.1 BradVision — Complete Camera & Vision System [Gen1]

BradVision is the perceptual nervous system — a vertically integrated stack spanning sensor optics, hardware ISP, AI computational photography, display integration, and security sensing.

```
Light → Optics → Sensor → ISP Hardware → BradNPU AI → SPMP → Display
```

5.2 Sensor Technology

Custom Quad-Bayer CMOS sensors co-engineered with BradLabs:

Spec	Lite	Standard	Pro	Elite
Resolution	48MP (12MP eff.)	108MP (27MP eff.)	200MP (50MP eff.)	200MP + 48MP dual-layer
Pixel size	1.0µm (2.0µm binned)	0.8µm (1.6µm)	0.64µm (1.28µm)	0.64µm + 1.0µm
Sensor size	1/2”	1/1.4”	1/1.12”	1” type

Spec	Lite	Standard	Pro	Elite
HDR	Staggered	Staggered + DCG	DCG + Multi-Sampling	Dual-gain + QBC
Readout	12-bit	14-bit	16-bit	16-bit dual-path
FPS	30fps full / 60fps 12MP	30fps / 120fps	24fps / 180fps	30fps / 240fps 50MP
Dynamic range	76dB	82dB	88dB	96dB

Pixel technologies: Dual Pixel PDAF (100%), Smart-ISO Pro (triple-gain), Staggered HDR, DCG HDR, QBC Re-mosaic (AI-trained), adaptive pixel-binning, dual-layer pixel (Elite only).

5.3 ISP Pipeline (14-Stage Hardware)

```
Sensor RAW → Black Level → Linearization → Defective Pixel → Crosstalk Comp
→ White Balance → AI Demosaic → Color Correction → Gamma → Tone Mapping
→ Edge Enhancement → Chroma Denoise → Lens Shading → Distortion Correct
→ Output (YUV/RGB/RAW) → SPMP
```

AI-Assisted Demosaic: Lightweight CNN trained on 10M+ RAW patches — 2× resolution recovery vs traditional binning, 80% false color reduction at high-ISO, <2ms per 48MP frame (dedicated CNN accelerator).

ISP Hardware Specs: 7nm dedicated ISP dielet, 200MP @ 30fps, 12 GPixel/s, 16 pixels/clock @ 800MHz, 4 TOPS AI accelerator, 8MB on-die SRAM, Fabric native (64 GB/s to SPMP), up to 4 concurrent streams.

5.4 Optics

Tier	Wide (Main)	Ultrawide	Telephoto	ToF/Depth
Lite	26mm f/1.8	13mm f/2.2	—	—
Standard	24mm f/1.7	12mm f/2.0	52mm f/2.4 (2×)	VCSEL ToF
Pro	23mm f/1.5	12mm f/1.8	70mm f/2.0 (3×) + 120mm f/2.8 (5×)	dToF
Elite	22mm f/1.4	10mm f/1.8	85mm f/1.8 (3.5×) + 170mm f/2.8 (7×) + 280mm f/3.5 (11× periscope)	dToF + IR flood

Technologies: 9-layer nano-coated AR, hybrid glass-polymer elements, BradSense piezoelectric micro-actuators (10µs response, 0.1µm precision), sensor-shift OIS (Elite, 5-axis, ±3°), liquid lens (Pro/Elite).

BradClarity UDC Display [Gen2]: Under-display camera with lower pixel density (200 PPI vs 460 PPI), transparent cathode (gallium-doped zinc oxide), AI compensation for diffraction pattern. Target: 32MP UDC indistinguishable from surrounding display.

5.5 BradVision Fusion Engine (Software on BradNPU)

Mode	Frames	Purpose	Compute
Smart HDR	3–7 staggered EV	Extended DR	2ms (16 NCC)
Night Fusion	8–15 short+long	Low-light recovery	5ms (32 NCC)
Action Fusion	4 ultra-short burst	Motion freeze + HDR	3ms (24 NCC)

Mode	Frames	Purpose	Compute
Portrait Fusion	2 wide+tele	Depth-aware isolation	4ms
Super-Res Fusion	4–9 micro-shifted	2×–3× resolution	8ms (64 NCC)
Focus Fusion	3–7 focus stack	Extended DoF (macro)	6ms (48 NCC)
Multi-Cam Fusion	All cameras	Unified optical zoom	10ms (64 NCC)

NeuroSense AI Weighting: Face priority (3× weight in portraits), edge detection, motion detection, texture analysis.

Continuum Flow: Capture session state preserves across Brad devices — start on phone, continue on slate, edit on BradMatrix Prime.

5.6 Video Capabilities

Tier	Max Resolution	Max FPS	Codecs
Lite	8K	30fps	H.265, AV1
Standard	8K	60fps	H.265, AV1, BradCompress
Pro	8K	120fps	H.265, AV1, BradCompress, ProRes RAW
Elite	8K	240fps	H.265, AV1, BradCompress, ProRes RAW, CinemaDNG

Features: Cinematic Mode (real-time DoF racking, adjustable after capture), Action Mode (4K120 with BradSense OIS + horizon lock), Night Video (temporal fusion 3-frame sliding window), Super Slow-Mo (720p960 / 1080p960, AI interpolation to 1920fps), BradCompress Video (~30% smaller than AV1, hardware decode in BradVision Engine).

5.7 Vision-as-Sensor

BradSec FaceLock: VCSEL + DOE (30,000 dots), 940nm IR flood, dedicated 2MP IR sensor → direct BradNPU, 3D face mesh, sub-100ms unlock, spoof detection (liveness + micro-expression + pulse on Elite).

BradVerify: Multimodal biometrics (face + iris + fingerprint), liveness challenge (random micro-expressions), on-device only.

XR / AR Interface: 6-DoF head tracking (1000Hz IMU + 60Hz camera), hand tracking (ultrawide 120° FOV, 60fps, 21-point skeleton), room mapping (ToF + UW real-time mesh via BradNPU), plane detection (<10ms per plane).

5.8 Modular Bayonet Optics [Vision]

Hot-swappable lens modules via magnetic bayonet mount: Portrait 50mm f/1.4 (28g), Macro 90mm f/2.8 (35g), Telephoto 200mm f/3.5 (65g), Wide 12mm f/2.0 (32g), Anamorphic 35mm f/1.8 (45g). Standardized MIPI connector + calibration EEPROM per module.

PART 6: PLATFORM & MOTHERBOARDS

6.1 SoC Platforms

BradMind — The Heart (Efficiency) [Gen1] **Philosophy:** “Smart, Cool, Enduring.” Optimized for performance-per-watt and silent operation.

Feature	Lite	Standard	Pro	Elite
CPU	P1+E4	P2+E4	P4+E4	P6+E4
GPU	BradFx 10 (8 VE)	BradFx 30 (16 VE)	BradFx 50 (24 VE)	BradFx 70 (32 VE)
NPU	256 NCC	384 NCC	512 NCC	768 NCC
Memory I/F	LPDDR5X	LPDDR6 / Brad-CAMM	Brad-CAMM DDR	Brad-CAMM LPDDR6
Process	7LPP (7nm)	SF4 (4nm)	SF4P (4nm)	SF4X (4nm)
TDP	2-5W	5-10W	10-25W	25-45W
Target	BradPhone Essential, wearables	BradPhone, BradSlate	BradSlate Prime, BradMatrix Essential	BradPhone Ultra, BradMatrix Essential

BradNeuro — The Brain (Performance) [Gen1–Gen2] **Philosophy:** “See, Sense, Simulate.” Sensory fusion at silicon level.

Feature	Ultra	Elite	X
CPU	P4+E8	P8+E4	BradXon (ANF)
GPU	BradFx 90 (48 VE)	BradFx 90 + BradGfx 100	BradGfx Max (ANF-integrated)
NPU	768 NCC + HB-AIM	1024 NCC + HB-AIM	X-Units (AIMP)
Fabric	Fabric + optical [Gen2]	Fabric + optical	Quantum-entangled [Vision]
Memory I/F	Brad-CAMM LPDDR6	Brad-CAMM LPDDR6	SPMP (LPDDR UMA)
Process	SF4X (4nm)	SF2 (2nm) [Gen2]	1.4nm→sub-1nm
TDP	45-65W	65-125W	125-350W+
Target	BradMatrix Prime, BradSlate Ultra	BradOS Compute WS	BradAstra headset (XR SKU), BradAstra servers, Brad-Skynet

6.2 BradBoard Motherboard Series [Gen1]

BradBoard Mobile — Ultra-Compact Embedded PCB, no user-accessible slots, SoC + memory soldered.

Variant	Size	Target	SoC	Memory
Mobile Nano	35×35mm	BradPhone Essential, wearables	Mind Lite	LPDDR5X
Mobile Standard	45×45mm	BradPhone Base, BradSlate Lite	Mind Standard	LPDDR6
Mobile Pro	55×55mm	BradSlate, BradMatrix Essential	Mind Pro	Brad-CAMM DDR
Mobile Elite	65×65mm	BradPhone Ultra, BradSlate Prime	Mind Elite / BradNeuro	Brad-CAMM

Construction: 8-12 layer PCB with embedded micro-fluidic channels, Direct-Path Trace Geometry (SoC-to-memory in microns), BradCool passive thermal, BradSec root-of-trust fused at fabrication.

BradBoard Matrix — Modular Plate Modular SoC socket + Brad-CAMM + Brad-M.3 slots for user-upgradable memory and storage.

Variant	Size	SoC	CAMM	M.3	Target
Matrix Air	150×100mm	Mind Pro/Elite	1× DDR	1×	BradMatrix Essential
Matrix Pro	180×120mm	Neuro Ultra	2× DDR	2×	BradMatrix Prime 14”
Matrix Studio	200×140mm	Neuro Elite	2× DDR	2×	BradMatrix Ultra 16”

Key technologies: SoC Module Socket (gold-plated compression pins, hot-swappable), Brad-CAMM flat screw-mounted modules, Brad-M.3 keyed slots, BradLightSpeed optical lane [Gen2], BradCool vapor chamber integrated into PCB stack.

BradBoard Nexus — E-ATX / Desktop Full-length expansion, dual-Xon support, complete Fabric implementation.

Variant	Size	SoC	CAMM	M.3	Target
Nexus Pro	305×244mm	Neuro Elite + 1× Xon	4× DDR	4×	BradOS Compute
Nexus Elite	305×267mm	2× Xon	8× DDR	6×	BradOS Compute Pro
Nexus Console	280×200mm (custom)	BradCore (8P+8E) + Gfx Max	2× GDDR7	2×	BradAxis

Key technologies: Dual SoC Socket [Gen1], SPMP Controller Hub, full-length BradGfx slot (x16 PCIe 6.0 + ANF direct-lane), BradLightSpeed Fabric Backplane [Gen2] (embedded optical waveguides, 12+ layer PCB), BradCool X-Liquid ports [Gen2], BradSec-X HSM soldered on-board.

BradBoard Rack — Data Center Composable compute sleds for BradSkynet and BradAstra servers.

Variant	Factor	CPUs	CAMM	M.3	GPUs	Target
Rack Sled	1U half	1× Xon	2× DDR	2×	—	BradSkynet node
Rack Quad	2U full	4× Xon	8× DDR	4×	4× Gfx Max	BradAstra server
Rack Matrix	4U	8× Xon	16× DDR	8×	N+1 redundant	Core cluster

6.3 Brad-CAMM Memory Slots

Feature	Brad-CAMM DDR	Brad-CAMM LPDDR6
Pin count	384	512
Data rate	8.8 Gbps (DDR6)	14.4 Gbps (LPDDR6)
Bus width	64-bit per module	128-bit per module
Form factor	Flat plate, screw-mounted	Flat plate + backplate
Thermal	Integrated heat spreader	Integrated heat spreader

6.4 Brad-M.3 Storage Slots

Custom 12mm-wide (vs M.2's 22mm), keyed for BradDrive only. Lite (x2, 32 GB/s, 42mm) and Pro (x4, 64 GB/s, 60mm).

6.5 BradLink Device Tethering [Gen2]

- **BradLink Local:** Wireless fabric extension (60 GHz, sub-mm wave) — phone offloads AI to nearby BradMatrix
- **BradLink Fabric Bridge:** Wired — two boards act as single shared-memory system
- **Latency:** <1µs local, <100µs tethered

6.6 Target Device → Platform Mapping

Device	SoC Platform	Motherboard	CPU	GPU
BradPhone Essential	Mind Lite	Mobile Nano	P1+E4	BradFx 10
BradPhone Base	Mind Standard	Mobile Standard	P2+E4	BradFx 30
BradPhone Ultra	Mind Elite	Mobile Elite	P6+E4	BradFx 70
BradSlate Base	Mind Standard	Mobile Standard	P2+E4	BradFx 30
BradSlate Prime	Mind Elite	Mobile Elite	P6+E4	BradFx 70
BradSlate Ultra	Neuro Ultra	Mobile Elite	P8+E4	BradFx 90
BradMatrix Essential	Mind Pro	Matrix Air	P4+E4	BradFx 50
BradMatrix Prime 14"	Neuro Ultra	Matrix Pro	P8+E4	BradFx 90
BradMatrix Ultra 16"	Neuro Elite	Matrix Studio	P8+E8	BradFx 90 + Gfx 500M

Device	SoC Platform	Motherboard	CPU	GPU
BradOS Compute	Neuro Elite	Nexus Pro	P8+E8	BradGfx 300
BradOS Compute Pro	BradXon	Nexus Elite	BradXon ×2	BradGfx Max
BradCube Base	BradChip Max	Matrix Studio-class	P8+E8	BradFx 90 (Max-i)
BradCube Prime	BradChip Max (bin)	Matrix Studio-class	P8+E8	BradFx 90 + eGPU lane
BradCube X Prime	BradXon (single)	Studio X-class	BradXon	BradGfx
BradCube X Ultra	BradXon (1–2)	Studio X-class	BradXon ×1–2	BradGfx Max
BradAxis Console	BradCore	Nexus Console	BradCore 8P+8E	BradGfx Max (fabric)
BradAstra Headset	Neuro X (XR)	Mobile Elite	P8+E8	BradFx 90 (integrated)
BradAstra Server	BradXon	Rack Quad	BradXon ×4	BradGfx Max ×4
BradSkynet Node	BradApex	Rack Sled	BradXon (controller)	— (AIMP focused)

PART 7: SOFTWARE & OS

7.1 BradOS Family [Gen1]

SKU	Target	Kernel	UI	Min RAM	Storage
BradOS Mobile	Smartphones	Hybrid (Zircon + Linux)	Touch-first, gesture	4 GB	64 GB
BradOS Slate	Tablets, foldables	Hybrid (Zircon + Linux)	Touch + Pen + Desktop	8 GB	128 GB
BradOS Compute	Laptops, desktops	Full Linux (mainline + Brad patches)	Desktop (KDE-based)	16 GB	256 GB
BradOS Nano	Wearables, IoT, BradTV	Zircon microkernel (no Linux)	Card-based, voice-first	512 MB	8 GB

Performance Tiers (Cross-SKU): - **Nano:** Ultra-light, smooth on 2GB RAM — comparable to Android Go - **Standard:** Balanced hero — comparable to Pixel OS meets Ubuntu - **Pro:** Unlocked, desktop-class multitasking, encrypted storage — comparable to macOS meets Fedora

Specialized Editions: - **BradOS Dev:** Compute + preloaded compilers, simulators, SDKs (BradCode Studio, BradCloud CLI) - **BradOS XR:** Compute x Nano — mixed reality for AR/VR headsets,

spatial UI, 6-DoF - **BradOS Gaming:** Compute Microkernel — BradAxis console OS, minimal overhead, instant-on - **BradOS Education:** Slate/Nano — lockdown mode, curriculum management, offline-capable

7.2 Core OS Features

NeuroSense AI Integration: - Predictive UI (app launch prediction based on time/location/context) - Contextual app launching (opening photo → camera shortcut) - Adaptive performance (EROE shifts power based on foreground task) - AI wallpaper (generative, adapts to time/weather/mood) - Smart notifications (priority sorting by learned behavior)

BradSec Integration: - Verified boot chain (PUF → bootROM → kernel signature) - Biometric unlock (FaceLock + fingerprint) - Per-file AES-256-GCM with keys in secure enclave - Zircon capability-based app sandboxing - Isolated secure enclave for biometric data

BradSuite Pre-installed Apps: BradBrowse (privacy-first WebKit), BradSearch (no profiling), BradMail (BradID + PGP), BradCloud (file sync, Continuum Flow-aware), BradOffice (local-first, BradCompress-native), BradPhotos (AI org, RAW editing), BradCode (terminal + IDE with BradDev Kit integration).

OTA Updates: Dual-slot A/B seamless updates, BradCompress-aware binary delta (~200 MB typical), firmware bundle updates (OS + Sec + Drive) in single reboot.

7.2.5 Software Support & Update Policy (Committed)

Brad Devices offers **long-lived software support** across every platform — a deliberate counter to planned obsolescence and a natural fit with the 4-year generation cadence.

OS & Firmware Updates — 7 Years Every device — **BradPhones, BradSlates, BradMatrix, BradOS Compute desktops, and the BradAxis console** — receives **7 years of OS and feature updates** from the product's launch date. Within those 7 years users get:

- **Major OS releases** (feature + architecture updates)
- **Security patches** (critical/zero-day fixes shipped promptly)
- **Stable, non-breaking updates** (never brick the device, always rollback-capable via A/B)

A 7-year window comfortably spans **~1.75 generations** of the 4-year cadence, so every device is covered for its own generation plus the majority of the next.

Platform	OS + Feature Updates	Notes
BradPhones / BradSlates	7 years	Matches/exceeds the flagship-phone norm
BradMatrix (laptops)	7 years	Laptop-class long-support
BradOS Compute (desktops)	7 years	Desktop long-support
BradAxis Console	7 years	Console-generation-spanning support

CPU & GPU Driver Support — 9 Years (Game-Ready) + 4-Year Legacy Graphics and compute drivers follow a **two-stage driver policy**:

1. **Game-Ready driver support — 9 years.** After a Brad Silicon CPU/GPU part launches, Brad ships **game-ready drivers for 9 years** — day-0 game optimizations, feature updates, and performance tuning for new titles and workloads.
2. **Legacy status — 4 further years.** After the 9-year game-ready window, the part enters **legacy status** for 4 additional years: security fixes, stability patches, and **bug-fix-only** updates (no new feature/game-day-0 optimization).

Net driver lifetime: **13 years** per Brad silicon part (9 game-ready + 4 legacy).

Stage	Duration	Includes
Game-Ready	years 0–9	Day-0 game optimizations, feature updates, perf tuning, new-workload support
Legacy	years 10–13	Security fixes, stability patches, bug-fix-only (no new features / day-0)

This policy is anchored in the **Kinetic and Torox driver layer** (one layer across all Brad silicon, `Brad Software/OS/BRADCVT_DRIVERS.md`), which makes long-lived, cross-SKU driver support structurally feasible. It is a core part of the "built to last, not built to be replaced" brand promise: a 4-year architecture that you'd otherwise feel pressured to swap is genuinely kept alive with software for a decade-plus.

Why Brad Can Sustain This — and Why It's a Deliberate Differentiator This is not charity and not a marketing line. Long-lived support is **structurally possible for Brad in a way it is not for the incumbents**:

- **No legacy code burden.** NVIDIA, AMD, and Intel carry decades of installed silicon, ISA baggage, and compatibility obligations. Every game-ready driver must be tested and tuned against hardware their engineers may no longer even have on a bench. Brad starts with **one clean driver architecture (the Kinetic and Torox driver layer) across a small, coherent silicon family** — so "optimizing for older parts" is the same effort as optimizing for new ones. We are not dragging a 20-year tail; we are tuning a single, young, well-understood stack.
- **No annual-release treadmill.** Because Brad ships a **4-year generation**, the engineering team is not forced to drop old parts the moment a new SKU launches. The cadence itself frees the headcount that incumbents burn on perpetual refresh. A slower hardware pace means every architecture and every driver gets *more* years of love, not fewer.
- **No corporate-bloat incentive to drop support.** Dropping software support early is how incumbents push new hardware and defend margins. Brad's model — fewer platforms, longer life, an upgradeable Brad-CAMM memory standard — does not depend on forcing churn. Obsolescence here is not a business lever; it is an accident to be avoided.

The result is a flywheel that builds **brand prestige and customer love**:

```
No legacy baggage + no annual refresh churn
        → engineers keep optimizing every part
        → old products stay genuinely good, not just "supported"
        → customers feel respected, upgrade on merit not pressure
        → brand trust compounds across generations
```

The **software policy is a deliberate differentiator**, not a compliance checkbox. It says: buy a Brad device and it will be made *better* over time, for longer than anyone else is willing to promise — because we are the only vendor whose architecture, cadence, and incentives are actually aligned to honor that promise. Where rivals treat "age of support" as a cost to minimize, Brad treats it as the core of the brand.

7.3 Brad Mobile Services (BMS) [Gen1]

BMS is the sovereign services layer — no dependency on Google, Apple, or any third-party.

Duality Strategy (launch): Ship with GMS as default for user familiarity. BMS as opt-in parallel. When ready — flip the switch.

Service	Description	Replaces
BradMail	Secure email with BradID	Gmail
BradHub	Curated app store	Play Store
BradPay	Fintech backbone, BradSec encrypted	Google Pay
BradConnect	Messaging + community platform	WhatsApp / Discord
BradCloud	Storage + sync	iCloud / Google Drive
BradBrowse & BradSearch	Privacy-first web + search	Chrome / Google Search
BradGeo	Mapping + location, smart city ready	Google Maps
BradNotify	Notification service (no Firebase)	Firebase Cloud Messaging
BradSphere	Social platform [Future]	Instagram / Facebook
BradPass	Premium subscription tier	Xbox Game Pass

Reference implementation (shipped): the REST v1 spec and a stdlib-only reference server ship at `src/bms/` (`api_v1.md` + `reference_server.py`, no third-party dependencies) — BradID auth, BradMail, BradHub, BradPay ledger (blake2b digests, 30-cent transfer fee), BradCloud, BradGeo, BradNotify. All 14 request/response cases verified against the running server; demo credentials and the error envelope are in the spec.

Privacy principles: Encryption by default, anonymization by architecture, local-first AI (all NeuroSense on-device BradNPU, not cloud), on-device consent, auditable open-source privacy client.

7.4 EROE — Eco-Render Optimization Engine [Gen1]

System-wide power and thermal governor embedded in BradOS.

Workload Profiles:

Profile	CPU	GPU	NPU	Memory	Example
Compute	80%	10%	10%	60%	Compiling
Gaming	30%	70%	20%	90%	Rise of the Streets
AI Inference	10%	10%	80%	70%	TIFA, BradVision
Video	10%	30%	0%	40%	AV1 decode
Idle	5%	0%	0%	10%	Lock screen
Battery Saver	30%	20%	20%	30%	Low battery

Cooling Integration: - <60°C: Max performance, no limits - 60-80°C: Gradual power cap, fan curve increase - 80-95°C: Hard limit 80% max power, aggressive fan - >95°C: Minimum throttle, display warning

API:

```
bradERoeSetProfile(EROE_PROFILE_GAMING);
bradERoeDeclareWorkload("render", EROE_UNIT_GPU, 0.7f);
```

Reference implementation (shipped): `src/drivers/eroe.c` implements the thermal governor for real — junction/board/battery telemetry, cooling thresholds, and a workload governor — exercised by the driver test suite and the BradDev Kit self-test.

7.5 Brad-CVT & Driver Suite [Gen1]

Philosophy: "Don't fight CUDA, absorb it." Every CUDA kernel, PyTorch model, game engine runs on Brad silicon without modification.

Brad-CVT (CUDA-to-Vector Translator)

```
CUDA Binary / PTX → Brad-CVT JIT/AOT (LLVM-based)
    → ISA Emulation Layer + Thread Remapper + Memory Redirector
    → BradVector-Optimized Kernel → BradGfx/BradFx
```

Components: - **AOT:** Offline compiler — CUDA PTX/SASS → BradVector ISA at install time - **JIT:** Runtime compiler — dynamically-generated kernels, sub-ms translation - **ISA Emulation Layer:** Dedicated silicon on every BradGfx/BradFx — translates legacy CUDA calls at wire speed - **Thread Remapper:** Maps CUDA warps (size 32) onto dual-issue BradVector lanes - **Memory Redirector:** CUDA shared → SRAM Infinity Buffer, global → SPMP LPDDR or HB-AIM

Supported CUDA Libraries (zero-modification): > Honesty note: these are *design targets* on the reference architecture, **not shipped-chip benchmark claims** — Brad silicon has not been built or measured against RTX 5090. Expected bandwidth/storage advantages drive the direction listed here, but real numbers await silicon. - cuBLAS: +15-30% vs RTX 5090 *target* (Infinity Buffer advantage) - cuDNN: +20-40% *target* (HB-AIM > HBM) - TensorRT: +25-50% *target* (MoE sparsity win) - NCCL: +30% *target* (on-die fabric > PCIe)

Native open standard support: SYCL 2020, Intel OneAPI, Vulkan 1.4, DirectX 12 Ultimate, Metal 3.

Kinetic & Torox Drivers The driver layer: one stack per architecture, together managing SPMP, AUC, Fabric, power, and memory across all Brad silicon.

Driver Modules: - **SPMP Manager:** Allocates LPDDR, zero-copy HDB→SPMP→Fabric, pool coalescing - **AUC Manager:** Manages 950MB MVRAM SRAM cache, cache-line granularity, TIFA AI prefetch - **Fabric Scheduler:** Gang scheduling across chiplets, NUMA-aware, optical lane steering - **Power Governor:** Per-cluster DVFS, 6 p-states, temperature-aware via BradCool telemetry - **Memory Allocator:** 4KB/64KB/2MB pages, huge page hints, zero-copy DMA - **Queue Manager:** Multi-queue (64 compute + 32 copy + 16 direct), HW priority lanes

Zero-Overhead Wrappers: Drop-in replacements for NVIDIA runtime — `#include <brad_runtime.h>` replaces `#include <cuda_runtime.h>`. No source changes needed.

Reference drivers (shipped): `src/drivers/` holds real reference implementations of the the driver layer's modules — SPMP first-fit pool allocator with coalescing + fragmentation stats, Fabric DMA/atomics/bind-memory, EROE thermal governor — and a new `bradlib` (`src/include/brad/bradlib.h`) exposes the platform as one C API: compile → pack → session → launch → run → inspect → breakpoints.

TIFA-SDK Developer tool that makes CUDA irrelevant for next-gen: - Python-first (`import tifa`), C++ backend - Auto-NeuroStream: model auto-partitioned across BradNeuro, BradVector, BradFx - MoE Spooler: expert prefetching to HB-AIM, cold spill to SPMP - One-line PyTorch: `model = tifa.optimize(model)` - FP8/FP4/INT4 native quantization, no calibration data needed

Reference package (shipped): `src/tifa/` is a working Python-first SDK — `import tifa`, AutoNeuroStream partitioner, MoE expert spooler, and the exact FP8 E4M3FN / E5M2 / FP4 E2M1 / INT4 digit formats (spot-checked bit-identical against the spec values: 240, 57344, 3.0, −5). Stdlib-only, zero dependencies.

7.6 BradDev Kit

Tool	Purpose
Brad-CVT CLI	`brad-cvt compile kernel.cu`

Tool	Purpose
brad-gpu	GPU monitoring (`info`, `top`)
Brad-Link API	C ABI (`bradLinkAlloc`, `bradLinkLaunch`, etc.)
Debugger	gdb-lldb hybrid with BradVector register inspection
Profiler	Per-kernel cycle/active/occupancy + memory timeline
TIFA Studio	GUI for model optimization and deployment
BFL	Brad Fusion Language — DSL for custom BradVector kernels (SYCL-based)

Reference implementations (released, `src/bradvector/`): `bradc` (basm → `.bvbc` assembler), BVRT (`.bvbc` interpreter), `bradgdb` (breakpoints, register/PC inspect), `bradtimeline` (per-cycle trace → CSV). These model the ISA at SIMT-VM level and run unattended in CI. See `Brad Software/OS/BRADVECTOR_PLATFORM_MOAT.md`.

The one-binary Kit (shipped): all of the above — plus the SPMP/EROE/Fabric drivers — are wrapped by the unified `braddev` CLI at `src/braddev/`:

Subcommand	What it really is
`braddev cvt`	bradc assembler (`.bvbs` → `.bvbc`, stdin OK, `-d` disasm)
`braddev run / dbg`	BVRT launch + bradgdb breakpoints; `--arg` seeds S16..S31; `--dump S#/M<addr>` scalar + memory inspection
`braddev gpu info / gpu top`	device profile + single-sample BradTimeline perf snapshot
`braddev timeline`	per-kernel cycle profile + CSV export
`braddev test`	self-test linking the drivers AND the whole toolchain pipeline

Verified end-to-end: SAXPY computes `y[i] = 4·x[i] + y[i]` over eight elements with zero traps (y[0] = 6.0, y[7] = 48.0 in float terms), fib(8) = 21, and all self-test checks pass. The reference ISA's honest rules are exercised in the examples: S0 is a read-only zero register, structural ALUs and `MOV` are float-domain, the integer ops (`IADD/ISUB/IMUL`) take registers only, and integer constants therefore enter via launch args — exactly the semantics silicon implements.

PART 8: CONSOLE & PRODUCTS

8.1 BradAxis Console [Gen1]

Target launch: Holiday 2027

Philosophy: Player-centric ecosystem — unrivaled performance, revolutionary immersion, sustainable ecosystem, accessibility.

Hardware

Component	Specification
CPU	BradCore console SKU (standard, 7LPP) — 8× Phoenix P + 8× Falcon E
GPU	BradGfx Max (fabric) — 1024 BradVector, 256 RT, 192 Neuro-Stream cores
NPU	BradNPU — 64 NCCs, 128 TOPS INT8
Memory	16/24 GB GDDR7 (CPU+GPU shared UMA) — ~1.0 TB/s base / ~1.15 TB/s Elite

Component	Specification
Storage	1/2 TB TLC BradNAND (SLC-cache), PCIe 6.0 x4, HDB (direct-to-GDDR)
Cooling	Vapor chamber + liquid metal TIM + active BradCool fan (whisper-quiet)
PSU	450 W internal, 80+ Platinum. Peak draw: 350 W. Standby: <1 W.

Connectivity: Wi-Fi 7, Bluetooth 6.0, 3× USB-C (front) + 2× USB-C (rear), HDMI 2.2 (8K@120Hz VRR), DP 2.0 (8K@120Hz DSC), 10 Gigabit Ethernet, 2× internal storage expansion.

Nexus Controller BradSense Adaptive Haptics: - 4× high-definition LRA actuators (20 Hz–10 kHz) - Texture feel (sand, ice, rain), environmental cues (tremors, wind), directional feedback, force/resistance

Dynamic Adaptive Triggers: - Programmable resistance, tension, tactile feedback - Variable pull weight (bowstring, weapon recoil), tactile clicks and hard stops

Biometric Feedback (opt-in): - PPG pulse sensor (right grip), GSR electrodes (both grips) - Game difficulty scaling based on stress/engagement - All data processed on-controller (RISC-V secure core) — raw data never leaves

NexusMods — Modular System: - Swappable D-pad (cross → disc → 8-way) - Swappable thumbsticks (convex → concave → extended precision) - Mini-OLED display (1.2” 480×480 — quick stats, inventory) - Precision trackball, custom macro pad (6 buttons), flight stick grip, back paddle expansion (2-4 paddles)

Wireless: BradLink proprietary (2.4 GHz + 5 GHz adaptive), <1ms latency, 2000 Hz polling, 15m range, 3000 mAh Li-Po (40h typical), USB-C fast charge (15min → 8h play).

BradOS Gaming Custom microkernel variant: - 3-second boot (power-on to game library) - <1 second game resume (instant state save/restore to GDDR) - Zero background processes — game gets 100% of CPU/GPU/NPU - Hardware-accelerated UI compositor (1% performance cost) - System updates applied during rest mode, never interrupt gameplay

Services

Tier	Price	Features
Free	$0	Online multiplayer, basic cloud saves, store access
BradAstra Unlimited	$15/mo	All first-party games day-one, 200+ library, cloud streaming, enhanced saves
BradPass	$25/mo	Unlimited + 7-day early access, $10/mo store credit, cosmetics, priority support, Experimental Labs

Developer Ecosystem: BradDev Kit (unified-memory-aware SDK — SPMP on mobile, GDDR on console; BradSense haptics, NexusMods API), BradLabs Program (funding for indie/emerging devs, esp. African), BradStudios Originals (sibling company under BradVerse Kenya — first-party titles, Rise of the Streets leads), 70/30 revenue split publishing.

8.2 Brad Ecosystem vs Apple Ecosystem (Short Form)

Full analysis: `BRAD_VS_APPLE.md`. Positioning (per the archive:83): *"the Apple + NVIDIA baby"* — Apple-style vertical integration + NVIDIA-style discrete rendering and open compute.

Layer	Apple	Brad
Silicon	A-series / M-series	HyperCore, BradChip/BradCore, BradXon
Memory	LPDDR UMA + 32MB SLC, soldered	SPMP 16–256GB + MVRAM 950MB + Brad-CAMM (upgradeable)
AI	Apple Intelligence, ~38 TOPS	NeuroSense AI, 32–128 TOPS, TIFA
GPU	Integrated only (MetalFX)	BradFx (integrated) + BradGfx (discrete, RT, NeRF)
Software/API	iOS/macOS, Metal only	BradOS; Vulkan 1.4 + DX12U + Metal 3 + SYCL; Brad-CVT (CUDA)
Services	iCloud, Apple Pay, Apple One	BMS: BradCloud, BradPay, BradPass
Display	Liquid Retina, ProMotion, XDR	BradLumin Display, SentinelView

Scorecard: - **Beats Apple:** memory flexibility (Brad-CAMM vs soldered), MVRAM freeze (~30× the system cache), bandwidth ceiling (1.84–2 TB/s), battery (36–48h), openness (4 GPU APIs + CUDA), discrete RT GPU tier, sovereign services. - **Trails Apple:** peak IPC (ROB 128 vs >150, lower frequency), node (7nm/4nm-class vs 3nm N3E), compiler maturity, software polish — and Apple ships while Brad is a spec. - **On par:** unified-memory philosophy, vertical integration, security posture (BradSec vs Secure Enclave).

PART 9: HORIZON & VISION

9.1 BradSkynet — AI Platform / AGI Infrastructure [Gen2]

Target launch: 2027-2028

Distributed AI compute platform running across the Brad ecosystem — from individual BradNPUs in edge devices to massive BradApex clusters in data centers.

Architecture:

```
BradSkynet Node
├── BradApex × 4–8       ← compute engine (see `Brad Silicon/Compute/BRADAPEX_ACCELERATOR_SPEC.md`)
├── BradXon CPU              ← node controller / orchestration
├── SPMP Enterprise (512 GB–1 TB pool)
├── BradDrive Enterprise (HDB direct-to-pool)
└── BradFusion Fabric (optical, sub-µs, inter-node)
```

BradApex (Gen1 design, Gen2 ramp 2027–28) replaced the original "BradGfx Max × N GPU cluster" concept as the node's primary compute — a ground-up datacenter accelerator (tensor engine for AI + first-class IEEE FP64 for HPC), memory-centric by design. BradGfx Max remains a graphics product; BradXon orchestrates but does not compute.

Proposed capabilities: Distributed MoE (experts sharded across nodes, sub-µs inter-node latency via optical Fabric), federated learning (edge NPUs train local models, gradient updates only to

Skynet), auto-scaling AI (workloads scale from phone to datacenter without re-partitioning), continuous learning (models update from live data streams).

9.2 BradAstra — VR Ecosystem & Headset [Gen2–Vision]

Target launch: 2030+ (Gen2 Synaptic window) — aligned to the 4-year generation cadence

BradAstra is Brad's **VR ecosystem and VR headset** — the answer to Apple Vision Pro / visionOS. Three layers under one brand: the **headset device** (BradNeuro X, BradOS XR), the **ecosystem** (persistent metaverse world, cross-device presence, creator economy), and the **infrastructure** (Brad-Skynet nodes / Rack Quad servers that render cloud scenes).

Headset: BradNeuro X (XR SKU, SF4X 4nm-class) mobile SoC — integrated BradFx Max-class + BradNPU; BradVision XR tracking (6-DoF head + hand, room mesh); 4K per-eye micro-OLED (UDC BradClarity); SPMP LPDDR unified pool; BradOS XR (spatial UI, hand tracking). Heavy scenes offload to the BradAstra server over BradLink/edge — mobile SoC + cloud server compute model.

Ecosystem features: Persistent world state in SPMP Enterprise streamed via BradCompress Video, neural semantic scene reconstruction, real-time AI NPCs (TIFA-powered), cross-device presence (Continuum Flow — walk in on the headset, continue on BradAxis), creator economy on BradDev Kit + BradLabs.

9.3 BradLabs — Fabrication & R&D Division

Current Role (2026) [Gen1]: - Process partnership with Samsung (unified across all Kinetic tiers — 7LPP→SF4X) — BradLabs provides custom PDK - BradNAND fabrication (238-layer 3D NAND) - Developer program: SDK preview, custom kernel workshops, expert weight trainer (NeRF training cloud tool) - Funding + mentorship for indie game developers (esp. African creators)

Future Vision (2029+) [Vision]: - Captive foundry: sub-1nm graphene/BN channels - BradNAND Gen2: 350-layer 3D NAND, 5+ GB/s per die - Optical interconnect: on-chip photonic communication - Bio-synaptic NPU: neuromorphic spiking neural networks - Ambient energy: BradWear powered by body heat + RF harvesting

9.4 Roadmap Chronology

Phase 1 — Foundations (Jul–Aug 2025): BradNPU, SPMP, BradGfx v1, BradXon original concept, BradOS lineup.

Phase 2 — Expansion (Sep 2025): BradRT Cores, BradVision, BradVerse CPU Lineup, BMS, BradGfx & BradFx Official Series, BradAxis Console, Additional Details.

Phase 3 — Refinement (Oct–Nov 2025): BradVision ISP, BradRAM, BradDrive + BradCompress, BradGfx v2.

Proposed Product Release Order: 1. Foundation IP: BradNPU, SPMP, BradCompress 2. Consumer Silicon: BradCore, BradFx, BradRAM, BradDrive 3. Platform Launch: BradOS, BMS, BradPhones/Slates 4. Performance Tier: BradGfx (Lite–Ultra), BradRT Cores, BradVision ISP 5. Flagship: BradAxis Console, BradGfx Max, BradAstra Unlimited 6. Apex: BradXon, BradSkynet, BradAstra, BradOS Compute Pro

9.5 Generation Timeline

Cadence: Every **4 years** Brad ships a new generation — a complete architectural leap, not an annual refresh. Within a generation, mid-cycle SKUs (node shrinks, memory upgrades, new form factors) refine the architecture without changing it.

Years	Gen	Name	Key Nodes
2026–29	Gen 1	Kinetic	BradMind/Neuro SoCs, BradBoard Mobile/Matrix/Nexus, Brad-CAMM, BradCool micro-fluidics, BradLink local (60 GHz)
2030–33	Gen 2	Synaptic	Node shrink (SF3/SF2 → 1.4nm), BradLightSpeed optical to Matrix tier, BradBoard Rack, BradLink Fabric Bridge (wired), BradSec-X to Neuro Elite, BradCool X-Liquid
2034–37	Gen 3	Quantum	Sub-1nm fabrication (vision), quantum-entangled Fabric to Nexus tier, graphene channel transistors, self-evolving logic, bio-synaptic NPU, ambient energy harvesting
2038–41	Gen 4	[Vision]	BradLabs captive fab scaling, deep on-die photonic logic, BN/2D-material dielectrics
2042+	Gen 5	Singularity	Self-evolving gates, quantum-entangled fabric, bio-synaptic

> **Note:** BradApex is a standalone datacenter accelerator, not a consumer generation — its [Gen1] design → [Gen2] ramp 2027–28 occurs *within* the Gen1 Kinetic window as a product ramp, complementing the 4-year consumer cadence.

PART 10: APPENDICES

Appendix A: Memory Hierarchy

```
REGISTER / L1 (per core) — sub-ns, KB
    ↓
L2 CACHE (per cluster) — ~1ns, hundreds of KB
    ↓
L3 CACHE (shared, on BradFusion Fabric) — ~5-10ns, MB
    ↓
L4 CACHE (on Brad memory-controller die — SRAM) — 2ns, 64 MB per stack
    ↓
SPMP (LPDDR Unified Pool — main memory) — ~20-40ns, 8-128 GB
    ↓
HDB L0 CACHE (on BradDrive Controller) — dictionary tables, recently decompressed
    ↓
BradDrive STORAGE (BradNAND) — PCIe 6.0, TB scale
    ↓
BradCloud / External (network, cold storage)
```

Appendix B: Security Boundaries

```
Layer 0: Fabrication (BradLabs) — hardware root of trust (PUF)
Layer 1: BradDrive Controller (BradSec) — encrypt at rest (AES-256-XTS)
Layer 2: HDB data path — transparent encryption going in, decrypted going out
Layer 3: SPMP (BradSec) — encrypt in memory (AES-256-GCM), integrity checks (Merkle)
```

```
Layer 4: BradRAM die (BradSec-RAM) — physical memory encryption per cache line
Layer 5: CPU/GPU (BradSec-X on Xon, BradSec on Core) — compute zone isolation
Layer 6: GPU Display Pipeline (BradSec at Pixel Level) — encrypted framebuffer, HDCP 2.3+
Layer 7: Software (BradOS, BMS) — user authentication, DRM, secure enclave
```

Appendix C: Data Pipeline Walkthrough — Game Level Load

```
1. Player triggers level load (or NeuroSense-Storage AI predicts it based on map position)
2. BradDrive Controller receives request → NeuroSense AI identifies required assets → BDC fetches from
3. Data passes through HDB: adaptive clocking ramps for burst → domain-specific decompression (geometry
to-SPMP pipeline
4. Data arrives in SPMP: NeuroSense AI Memory Controller organizes by priority (render > AI > game logic
5. Processors access via BradFusion Fabric: GPU reads textures/meshes/shader, CPU reads game state/phy
6. BradRAM DDS learns access patterns → prefetches next likely assets → caches in L4 SRAM → loop continu
```

Result: Loading screens eliminated. Zero CPU tax on any data path.

Appendix D: Bandwidth Budget

Stage	Bandwidth	Notes
BradDrive → HDB	64 GB/s	Raw NAND (PCIe 6.0 x4 uni)
HDB decompression	Multiplies by 2-4×	Hardware-accelerated, no CPU cost
HDB → SPMP	Same, zero-copy	Direct pipeline
SPMP → Fabric	Exceeds LPDDR standard	3D-stacked TSVs reduce distance
Fabric → Processor	Sub-ns per hop	Mesh NoC topology

Appendix E: Naming Conventions

Correct Term	Wrong Term(s)	Type
BradFusion Fabric	BradFusion Platform & Chipset	On-chip mesh NoC
BradFusion Platform	—	Superset (Fabric + I/O + controllers)
BradFx Standard	Balanced	GPU tier naming
BradSlates	BradPads	Tablet product name
BradCompress HDB	Hardware Decompression block / BradCompress Decompression Block	Full formal name
BradGfx Max (Discrete)	—	Standalone card
BradGfx Max (Integrated)	—	In BradXon via ANF
BradGfx Max Xntensive	—	Discrete GDDR7 hybrid-UMA edition (192–512-bit bus)
BradSec	—	Base hardware security

Correct Term	Wrong Term(s)	Type
BradSec-RAM	—	Inline memory encryption in the memory-controller data path (AMD-SME/MKTME-class, protecting standard JEDEC dies)
BradSec-X	—	Quantum-fortified, self-healing
BradSec Display Pipeline	—	Pixel-level content protection
SPMP	—	Logical unified pool from BradRAM dies + controller
BradRAM	—	Physical LPDDR5X die technology powering SPMP

Appendix F: Credibility Tiers

Grounded [Gen1] — real products, real engineering: BradNPU, BradRAM, SPMP, BradDrive, BradCompress, BradVision ISP, BradAxis Console, BradOS, BMS, EROE, BradCore, BradFx/BradGfx (excluding 2GB SRAM), BradRT Cores, HyperCore/Atlas, BradChip, Brad-CVT/Drivers

Mixed — real core + visionary extensions: BradXon (ANF is [Gen1] — compute partitions, RDA, fabric governor; the unit-morph is [Vision]), BradGfx Max 2GB SRAM Infinity-Buffer ([Vision] — 3D-stacked tile under compute die), BradSense-Render (core is [Gen1], full neural rendering is [Vision]), BradClarity UDC ([Gen2]), BradFusion Fabric optical interconnects ([Gen2]), SPMP LPDDR UMA ([Gen2])

Visionary [Vision] — requires fundamental breakthroughs: Sub-1nm graphene channel transistors, quantum-entangled Fabric, self-evolving logic gates, bio-synaptic NPU, ambient energy harvesting, 2GB monolithic SRAM, on-die photonic logic

Aesthetic — marketing language, not technical: “Living compute substrate,” “hardware embodiment of consciousness,” “synthesize reality,” “beyond von Neumann,” “Sovereign Visuals,” “soul of sight”

Appendix G: Open Questions for Brad

1. **BradBoard vs chassis integration:** Should high-end Matrix/Nexus boards integrate into chassis or remain swappable? The archive suggests “cable-less” is the future direction.
2. **Brad-CAMM standard:** Lock in pin count and keying for Gen 1? Need to ensure LPDDR6 14.4+ Gbps signal integrity.
3. **BradLink security model:** When a BradPhone tethers to a BradMatrix, should the laptop trust the Phone’s BradSec, or require independent authentication?
4. **Rack sled form factor:** Standard 1U half-width, or custom BradLabs width for better thermals?

5. **Nexus Console vs standard Nexus:** Should BradAxis use a custom board or modified Nexus Pro with Xon-only? **Resolved:** custom 280×200mm board, soldered BradCore 8P+8E + GDDR7 (no Xon involvement).
6. **Full custom kernel timeline:** BradOS 3.0 with BradKernel (no Linux dependency) targeted for 2028. Realistic?

Appendix H: Key Architectural Relationships

1. **Data Pipeline (the real differentiator):** `BradDrive → BradCompress HDB → SPMP → BradFusion Fabric → any processor`. Zero-copy, zero-CPU-overhead, direct-to-pool streaming.
2. **BradFusion Fabric:** Mesh NoC interconnecting all components at hardware level. Extends from die → package → board → rack (optical in [Gen2], quantum-entangled in [Vision]).
3. **SPMP is the nexus:** Every compute element pulls from the same unified LPDDR pool via the Fabric.
4. **CPU–GPU integration:** BradCore (workhorse) has integrated BradFx. BradXon (flagship) has integrated BradGfx Max via ANF. BradGfx Max Xntensive is the discrete GDDR7 edition (hybrid-UMA).
5. **HyperCore – BradCore relationship:** HyperCore/Atlas is the mobile specialization — context persistence over throughput. Same shader math, same architecture patterns, different optimization target.
6. **BradRAM → SPMP relationship:** BradRAM is the physical LPDDR die. SPMP is the logical pool. Multiple BradRAM dies + controller = SPMP. They are the same physical silicon at different abstraction layers.

Appendix I: Rise of the Streets

The ecosystem's showcase reference game. **Still in development** — no dedicated documentation (beyond SDK references in `Brad Software/OS/BRADCVT_DRIVERS.md`). Referenced in BradAxis, BradDrive, and BradSense-Render docs as the title that demonstrates the pipeline — seamless open-world streaming, no loading screens, NeuroSense-AI-predicted asset pre-fetching, per-object neural reconstruction. A reference title, not the product itself. Mobile-compat: same shader math on HyperCore, runs on BradSlate without a port.

Appendix J: Complete Component Index

Component	Type	Badge	Defined In
BradCore	CPU	[Gen1]	BRADCHIP.md
BradXon	CPU (Flagship)	[Gen1]; ANF unit-morph [Vision]	PLATFORM_SPEC.md
HyperCore HC-100/300/500/Elite	SoC (Mobile)	[Gen1]	HYPERCORE_ATLAS.md
Atlas Architecture (AUC, Sentinel, MVRAM)	Architecture	[Gen1]	HYPERCORE_ATLAS.md
BradChip Lite→Max	SoC Package	[Gen1]	BRADCHIP.md
BradGfx Lite→Max	GPU (Dedicated)	[Gen1]	GPU_ARCH.md
BradFx Lite→Max	GPU (Integrated)	[Gen1]	GPU_ARCH.md
BradVector ISA + Engines	GPU Vector ISA/Compute	[Gen1]	GPU_ISA.md, GPU_COMPUTE.md
BradRT Cores	RT Accelerator	[Gen1]	GPU_ARCH.md

Component	Type	Badge	Defined In
BradSense-Render	Neural Rendering	[Gen1]	BRADSENSE_RENDER_SPEC.md
Neuro-Stream Units	GPU AI Cores	[Gen1]	BRADSENSE_RENDER_SPEC.md
BradNPU (+NCC, HB-AIM, MoE, AI-ISA)	NPU	[Gen1]	BRADNPU_ARCH.md
BradRAM (+L4 SRAM)	Memory Die	[Gen1]	BRADRAM_SPMP_ARCH.md, BRADRAM_DDS.md/BradRAM_POWER/R
SPMP (Lite→Enterprise)	Unified Pool	[Gen1]	BRADRAM_SPMP_ARCH.md, BRADRAM_VMEM.md
NeuroSense AI Memory Controller	Controller	[Gen1]	BRADRAM_SPMP_ARCH.md
BradSec-RAM	Security (Memory)	[Gen1]	BRADRAM_SPMP_ARCH.md, BRADRAM_RAS.md
BradDrive (Standard/Pro/Elite/Enterprise)	SSD	[Gen1]	BRADDRIVE_COMPRESS_ARCH.md
BradCompress (+HDB)	Codec + HW	[Gen1]	BRADDRIVE_COMPRESS_ARCH.md
BradVision (Lite→Elite, ISP, Fusion, Optics, Coach)	Camera/Vision	[Gen1]	BRADVISION_COMPLETE.md
BradOS (Mobile/Slate/Compute/Nano + Editions)	OS	[Gen1]	BRADOS_COMPLETE.md
BMS (10 pillars)	Cloud Services	[Gen1]	BMS_SPEC.md, BRADOS_COMPLETE.md
EROE	Power Governor	[Gen1]	EROE.md, BRADOS_COMPLETE.md
Brad-CVT (+ AOT/JIT, ISA Emulation)	CUDA Translator	[Gen1]	BRADCVT_DRIVERS.md
Kinetic & Torox Drivers	Driver Stack	[Gen1]	BRADCVT_DRIVERS.md
TIFA-SDK	AI SDK	[Gen1]	BRADCVT_DRIVERS.md
BradVector Platform (bradc/BVRT/bradgdb)	SW Toolchain (reference)	[Gen1 ref]	BRADVECTOR_PLATFORM_MOAT.md, GPU_ISA.md
BradAxis Console (+Nexus Controller, NexusMods)	Console	[Gen1]	BRADAXIS_SPEC.md
BradMind Lite→Elite	SoC Platform	[Gen1]	PLATFORM_SPEC.md
BradNeuro Ultra→Elite/X	SoC Platform	[Gen1–Vision]	PLATFORM_SPEC.md
BradBoard Mobile/Matrix/Nexus/Rack	Motherboard	[Gen1]	PLATFORM_SPEC.md
Brad-CAMM LPDDR/DDR	Memory Module	[Gen1]	PLATFORM_SPEC.md
Brad-M.3	SSD Form Factor	[Gen1]	PLATFORM_SPEC.md
BradCool (micro-fluidics, X-Liquid, Nano)	Cooling	[Gen1–Gen2]	PLATFORM_SPEC.md
BradLink (Local, Fabric Bridge)	Tethering	[Gen2]	PLATFORM_SPEC.md
BradSkynet	AI Infrastructure	[Gen2]	BRADSKYNET_ASTRA_LABS.md
BradAstra	VR Ecosystem + Headset	[Gen2–Vision]	BRADSKYNET_ASTRA_LABS.md
BradLabs	Fab/R&D	[Gen1–Vision]	BRADSKYNET_ASTRA_LABS.md
BradClarity UDC	Display	[Gen2]	BRADVISION_COMPLETE.md
Modular Bayonet Optics	Lens System	[Vision]	BRADVISION_COMPLETE.md
BradLightSpeed (hybrid photonic)	Interconnect	[Gen2]	PLATFORM_SPEC.md
Quantum-entangled Fabric	Interconnect	[Vision]	ARCHITECTURE.md

Component	Type	Badge	Defined In
Self-evolving logic gates	CPU Tech	[Vision]	VISION.md
Bio-synaptic NPU	NPU Tech	[Vision]	VISION.md
Ambient energy harvesting	Power	[Vision]	VISION.md
BradSec-X Quantum Fortification	Security	[Vision]	VISION.md

End of Brad Devices Complete Unified Specification

Compiled from 78 supplementary `.md` reference documents + a 2,167-line AI co-architect conversation (proprietary). Total ecosystem: 84 text files, 17,635 lines (excluding Converted/ originals; 86 files, 18,527 lines with them). This compendium: 1,738 lines, 41 pages (merged/deduplicated from the supplementary docs). Two directories are classified [ARCHIVE] and are excluded from the 79: `Converted/` (17 pre-conversion source exports + `ARCHIVE.txt`) and `Peripherals/[proprietary]` (2,167-line raw co-architect transcript, + `Peripherals/ARCHIVE.txt`). Both keep superseded framing on purpose — see each directory's `ARCHIVE.txt` and the archive manifest. Never cite an archive; cite the live doc. Created by The Architect, Jul–Nov 2025. Consolidated Jun–Jul 2026. Refreshed Sept 2026. "Data in Motion. Intelligence in Real-Time." — Kinetic (Gen 1)